

Supplemental Material: Coupling-gated cortical coherence

Overview

This document supplies the mathematical constructions and experimental methods for the three requirements developed in the main article: coherence, compatibility and self-representation. It is organized as a technical reference, beginning with supporting thermodynamic constraints and the formal constructions, followed by physical calibration, recovery tests and measurement specifications. The thermodynamic results constrain a proposed implementation; their order does not derive the cortical hypothesis from finite capacity.

For the physical test, the finite mean-field controls and the recovery-mechanism construction give the full evidence behind the article’s calibration and identifiability argument. For compatibility, the gluing construction states the overlap assumptions, and the joint-plasticity sections report all control arms, alignment statistics and complete-interval measurements. For self-representation, the reflexive construction distinguishes fixed-point uniqueness from the additional condition that encoding is restriction. Table S1 maps the mathematical claims to their formal statements. Source code and saved summaries are maintained at https://github.com/bobaseb/coupling_gated_cortical_coherence.

1 Symmetry Breaking and the Emergence of Boundaries

A boundary is one of the physical structures the broader composition assumes. Under the stated vacuum and connectedness hypotheses, a field reaching two vacuum components must leave the vacuum somewhere. Relating that locus to a finite information-processing register is a separate physical premise; the result supplies neither a general condition for agency nor a derivation of the cortical coupling model.

Let $\mathcal{M}$ be a pseudo-Riemannian manifold representing spacetime, and let $\phi : \mathcal{M} \rightarrow \mathcal{V}$ be a physical field mapping to a vacuum manifold $\mathcal{V}$ (the space of energy minima). Let G be a continuous symmetry group acting on $\mathcal{V}$.

Theorem 1 (Domain-wall obstruction): A continuous field on a connected substrate whose image reaches two different connected components of the vacuum manifold must leave that manifold somewhere. This is the π_0 case of the topological-defect argument. A separate extension obstruction states that non-null-homotopic boundary data admit no continuous extension over the specified contractible interior.

Lean 4 Implementation (Phase1_Primitives.lean): We formalize this by defining the set of degenerate energy minima `DynamicalVacuum` and proving `boundary_defect_forces_interior_vacuum_break`: if a boundary field configuration ϕ_∂ is not null-homotopic, then it admits no continuous extension over a contractible interior. This is an obstruction to extending the specified boundary data. Note the direction: this result establishes that a defect *obstructs extension*, not that a defect must exist. The companion lemma `contractible_interior_forces_trivial_boundary` proves the contrapositive.

A direction that produces a defect is proved, at the level of π_0 . `exists_notMem_of_no_common_preconnected` states it in the form a witness can discharge — if no preconnected subset of the vacuum manifold M contains both $\phi(x_1)$ and $\phi(x_2)$, and the substrate is connected, then ϕ leaves M somewhere — and `exists_notMem_of_connectedComponentIn` restates it with $\phi(x_2) \notin$ the connected component of M containing $\phi(x_1)$. The proof uses the connectedness of continuous images and `IsPreconnected.subset_connectedComponentIn`, and nothing else; no hypothesis mentions energy, a potential or a symmetry. This is the domain-wall case of Kibble’s classification and we take only that case: $\pi_1 \neq 0$ forcing vortices and $\pi_2 \neq 0$ forcing monopoles are recorded as available and not taken, since the manuscript’s argument uses domain walls alone.

Separately, `spontaneous_symmetry_breaking` shows that a global energy minimiser lies in the vacuum manifold almost everywhere. The proof introduces no additional axioms and uses `pointwise_vacuum_of_global_min`.

Examples §11 witnesses that result on the double well $V(v) = (v^2 - 1)^2$ over a two-point spacetime carrying a Dirac probability measure. Two features of the witness are deliberate. Its minimisation hypothesis is *proved* rather than assumed — everywhere else in the development the existence of a global minimiser is a hypothesis, since no compactness or direct-method argument is formalized. And the almost-everywhere in the conclusion is shown necessary: the minimising field sits at the vacuum on the support and at 0 off it, so the pointwise version of the statement is false of the very field the theorem applies to. The same section instantiates the $\mathbb{Z}_2$ action $v \mapsto \pm v$ and discharges the energy invariance, giving the first instance of `SymmetryInvariantAction`; the vacuum manifold $\{-1, 1\}$ is proved degenerate and the two minima exchanged by the symmetry, so a minimiser and its image have equal energy and differ.

Examples §11.1 then supplies the witness for the π_0 theorem on the same well. The vacuum manifold is *computed* there, `wellVacuum_eq` proving `DynamicalVacuum` $V = \{-1, 1\}$ exactly rather than merely containing both minima — which matters, because a larger vacuum set might have been connected — and `wellVacuum_separated` proves the disconnection in the form the theorem consumes, by the observation that a preconnected subset of $\mathbb{R}$ spanning -1 and 1 contains 0 , which is not a minimum. `wellV_domain_wall` is the resulting statement about *every* continuous field on *any* connected substrate reaching both minima. Non-emptiness of that class is checked separately rather than assumed: the clipped identity on $\mathbb{R}$ satisfies the hypotheses, and `kink_zero_notMem` locates its forced wall at the origin. These results concern vacuum membership and domain walls; they construct no conserved quantity.

2 The Thermodynamic Constraint: Landauer’s Principle

For a finite effective register, the number of distinguishable histories that can be retained is bounded. This does not imply that every update erases. The capacity theorem states that a distribution on a finite state space X carries at most $\log |X|$ nats, with equality only at the uniform law (`shannon_entropy_eq_log_card_iff`), and a perturbation stream of length k over an alphabet P with $|P|^k > |X|$ therefore cannot be recorded injectively (`absorb_not_injective`), whatever the update map is and even when every individual step of it is reversible.

Let the global phase space $\Omega = \Omega_{sys} \times \Omega_{bath}$ undergo globally reversible Hamiltonian dynamics, modeled as an injective mapping $f : \Omega \rightarrow \Omega$.

Theorem 2 (Thermodynamic Erasure): Under the specified finite-register and reversible system–bath assumptions, the heat identity gives a lower bound proportional to the entropy erased by the register. Erasing one unbiased bit gives $\Delta Q \geq k_B T \ln 2$. Non-injectivity alone does not specify a universal one-bit entropy loss; the bound uses the erased entropy of the particular update.

The same heat, in three vocabularies (Phase3.LandauerBridge.lean). Derivation 2 charges a register for destroying information and Derivation 3 charges a structure for keeping information it cannot use. `PredictiveDissipation.ofLandauer` builds the predictive structure of a given register: its thermal scale is that register’s temperature, its dissipated work is that register’s Landauer heat, and Still’s bound is derived from `landauer_bound` rather than assumed a second time. The single hypothesis is the identification $(I_{mem} - I_{pred}) \leq S(id) - S(t)$ — the memory wasted is no larger than the entropy erased — and it is a restriction, not bookkeeping: on an injective update the right-hand side is zero and the inequality forces the waste to vanish (`nonpredictive_eq_zero_of_injective`), which is the Norton–Shenker distinction in informational form. The measure-theoretic statement of Landauer (`Phase3.MeasureThermodynamics`) is specialised in the same file rather than left standing beside the combinatorial one: at counting measure on a finite register, `continuous_entropy` is `boltzmann_entropy`, volume contraction is non-injectivity, and `landauer_heat_bound` is T times the entropy erased (`landauer_heat_bound_count_eq`). The witness is the one-bit eraser, where a register destroying $\log 2$ pays for a memory wasting $\log 2$, with equality (**Examples §18.5**).

Lean 4 Implementation (Phase3.CombinatorialThermodynamics.lean): We explicitly decouple logical erasure from physical erasure. Lean verifies that any contraction in the *physical* state space ($\neg \text{Function.Injective}(t_{sys})$) under globally reversible dynamics mathematically necessitates heat dissipation. The combinatorial core, `landauer_from_reversibility` and `second_law`, is fully derived. Finiteness of the phase space is what converts the physically checkable hypothesis into the one the bound needs: on a finite state space a map that fails to reach every state is not injective (`is_erasure_of_not_surjective`), so an unreachable state suffices for $\Delta Q > 0$ (`finite_phase_space_dissipates`). The hypothesis is fenced: `Nat.succ` misses 0 and is injective, so the implication is false on an infinite phase space, and **Examples §19** records this beside the witnesses. The Norton–Shenker distinction survives the identification — retaining the incoming perturbation is injective and free (`keepRecord_injective`), refreshing

the register that held it is not (`absorbStep.isErasure`). Converting it into a statement about heat requires the postulate $\Delta Q = T\Delta S_{\text{bath}}$, which is carried as the class field `StatisticalMechanics.heatEq`; `Examples §1` exhibits a one-bit erasure system with an explicit bath that discharges it, so the postulate is demonstrably satisfiable. A register that satisfies the stated erasure and thermal hypotheses dissipates; identifying it with a biological boundary is an additional modelling step.

3 Structural Resonance and Least Action

A bound relating environmental and internal distributions must be a physical assumption about specified processes, not a consequence of Gibbs’ inequality alone. The class field `StructuralResonance.kl_bound` states $D_{KL}(P||Q) \leq \Delta t \sigma$ for an instance. Even under this assumption, descent of σ does not imply $D_{KL} \rightarrow 0$ unless the upper bound vanishes. A positive limiting upper bound permits a positive divergence. No convergence of internal geometry to environmental structure follows.

Theorem 3 (Nonpredictive information bounds dissipation): Let X_t be the system’s state and S_t an environmental signal evolving by a dynamics of its own on which the system does not act. Then

$$I(X_t; S_{t+1}) \leq I(X_t; S_t), \quad W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})], \quad (1)$$

the second being the bound of Still et al. (2012). Consequently $W_{\text{diss}} \geq 0$, and any bound on the dissipated work is a bound on the nonpredictive information $I(X_t; S_t) - I(X_t; S_{t+1})$.

The first inequality is the data processing inequality for the Markov chain $X_t \rightarrow S_t \rightarrow S_{t+1}$, proved in `Phase3.PredictiveThermodynamics.lean` as `predictiveInfo.le_mutualInfo`; mutual information is defined there from Mathlib’s `klDiv` as the divergence of the joint law from the product of its marginals, since Mathlib carries no mutual information of its own. The second inequality is the class field `PredictiveDissipation.still_bound`, an irreducible physical postulate for the same reason as every other one here — its `axiom` form is refutable, and `Examples §18.4` refutes it. Everything downstream is derived: non-negativity of the dissipated work, the bound of the nonpredictive information by the dissipation budget, and the statement that a quasi-static drive leaves every retained bit predictive. `Examples §18` inhabits the class twice over one correlated two-bit law — a frozen signal, permitted to dissipate nothing, and a scrambled one, forced to dissipate its entire memory — and fences the theorem with a Markov kernel on the pair that writes the system’s state into the signal and thereby *raises* mutual information, so the no-back-action hypothesis is shown to carry weight rather than to decorate. The equality of Kawai et al. (2007), $\langle W_{\text{diss}} \rangle = k_B T D_{KL}(\text{forward} || \text{reverse})$, is carried as a predicate on an instance and yields `nonpredictive.le_arrow`: the irreversibility of the dynamics is at least the memory that fails to earn its keep.

3.1 Agency and the scope of the passive bound

The stochastic channels of `Agency (Phase3.Agency.lean)` specify one perception–action cycle:

$$\begin{aligned} A_t &\sim \pi(\cdot | X_t), & S_{t+1} &\sim K(\cdot | S_t, A_t), \\ O_{t+1} &\sim G(\cdot | S_{t+1}), & X_{t+1} &\sim U(\cdot | X_t, A_t, O_{t+1}). \end{aligned}$$

The sampled action is retained through actuation and supplied to the memory update. The history used for predictive information records (X_t, S_t, S_{t+1}) , keeping memory at its pre-action state. Its present marginal is the input law and its future marginal is the post-action world paired with that pre-action memory. The environmental state must be sufficient for the declared Markov dynamics; an observation alone need not satisfy this condition.

`Feedback.information_balance` proves Eq. (10) from the existing KL-based mutual information. Conditional information uses the chain-rule expression $I(X; (S, S')) - I(X; S')$ or $I(X; (S, S')) - I(X; S)$, respectively. Data processing proves both nonnegative under finite joint information, and `mutualInfo.finite` discharges that condition on finite discrete spaces, including laws with zero atoms. Real subtraction preserves the sign of their difference. The passive constructor retains the present signal while evolving it through its own kernel; `passive_feedback_zero` and `passive_signedWaste` recover the passive information statements exactly. The witness `writingAgent` selects its own bit as the action, writes it into an initially independent world bit, senses the result and updates its memory. Its signed difference is strictly negative, and its composed cycle equals the explicit write kernel. Conditional dependence identifies causal influence only with an adequate causal model.

3.2 A finite thermodynamic model with feedback

For a finite controller–world law $p(x, s)$ and a normalized channel $K_x(s' | s)$, one elementary update holds x fixed. The final law and the forward and reverse path laws are

$$\begin{aligned} p'(x, s') &= \sum_s p(x, s) K_x(s' | s), \\ P_F(x, s, s') &= p(x, s) K_x(s' | s), \\ P_R(x, s, s') &= p'(x, s') K_x(s | s'). \end{aligned}$$

Both path laws are normalized. The reverse uses the same autonomous transition channel, started at the forward final distribution. The theorem assumes strictly positive initial masses and transition probabilities. Time-dependent protocols and unidirectional transitions require a corresponding treatment of reversal and support.

`FiniteFeedbackStep.entropy_balance` (`Phase3AgencyThermodynamics.lean`) expands the path log-ratio to obtain

$$\Sigma = D_{KL}(P_F \| P_R) = H(p') - H(p) + \sum_{x, s, s'} P_F(x, s, s') \log \frac{K_x(s' | s)}{K_x(s | s')} \geq 0. \quad (2)$$

Gibbs’ inequality proves nonnegativity; the balance is derived from these distributions. `ProbDist.KL_eq_klDiv` identifies the finite divergence with Mathlib’s measure-theoretic divergence for a positive reference law. The information identity uses the same forward law and its proven initial and final marginals.

Local detailed balance is the physical identification $q(x, s, s') = k_B T \log[K_x(s' | s)/K_x(s | s')]$, with heat positive into the reservoir Horowitz and Esposito (2014); Ito and Sagawa (2013). It gives $\langle Q \rangle \geq k_B T [H(p) - H(p')]$. The first-law hypothesis $w = E(x, s') - E(x, s) + q$ separately fixes work. These are bounds on the joint controller–world system. The passive signed information difference alone is insufficient to supply a general work bound. `Chain.active_entropy_budget` gives the scalar budget consequence of `ActiveBound`, the fourth node of the active composition.

Complete finite perception–action update. `FiniteFeedbackCycle`, in the same module, carries an actuation step and a memory channel $U_t(y | x)$. Its sensing step starts at $p_1^{\text{swap}}(t, x) = p_1(x, t)$, where

$$p_1(x, t) = \sum_s p_0(x, s) K_x(t | s), \quad p_2(y, t) = \sum_x p_1(x, t) U_t(y | x). \quad (3)$$

The second initial law is constructed from the first final law. The theorem `cycle_law` identifies p_2 with the output of the composed `Agency.cycle` channels, with action $a = x$ and observation $o = t$. Strictly positive p_0, K, U give positive substep path laws. At a common thermal scale $\theta = k_B T > 0$, local detailed balance for each channel yields

$$\Sigma_A + \Sigma_M = H(p_2) - H(p_0) + \frac{\langle Q_A \rangle + \langle Q_M \rangle}{\theta} \geq 0. \quad (4)$$

The intermediate entropy cancels because exchanging coordinates preserves joint entropy. No intermediate or final law is assumed stationary. `entropy_budget` therefore gives $H(p_0) - H(p_2) \leq B/\theta$ whenever $\langle Q_A \rangle + \langle Q_M \rangle \leq B$.

With one energy observable $E(x, s)$, the separate first-law hypotheses are $w_A(x, s, t) = E(x, t) - E(x, s) + q_A(x, s, t)$ and $w_M(t, x, y) = E(y, t) - E(x, t) + q_M(t, x, y)$. `mean.first_law` cancels the same intermediate expected energy and proves

$$\langle W_A \rangle + \langle W_M \rangle = \mathbb{E}_{p_2} E - \mathbb{E}_{p_0} E + \langle Q_A \rangle + \langle Q_M \rangle. \quad (5)$$

These results cover the two autonomous channel substeps with action and observation aliases. Extra physical registers, time-dependent control and learning require specified dynamics and costs. These balances supply no upper budget; the register ledger below allocates one.

Reciprocal two-bit witness. In units $k_B T = 1$, let the interaction energy be zero for $x = s$ and log 3 for $x \neq s$. Each substep redraws one bit, agreeing with the fixed other bit with probability 3/4. Actuation updates s using action $a = x$; sensing updates x using observation $o = s$. The action and observation are aliases of existing states and require no separate storage register. Their physical content is the specified pair interaction and thermal updates. The alignment preference is supplied by that interaction.

Starting from independent uniform bits, actuation produces probability $3/8$ on each agreeing pair and $1/8$ on each disagreeing pair. Both marginals remain uniform. The subsequent sensing step preserves this joint law while allowing individual memory transitions. In `Examples/AgencyThermodynamics.lean`, the exact actuation heat is $(\log 3)/4$ and entropy production is $\log 2 - (\log 3)/2 > 0$; the sensing step has zero mean heat and entropy production. Each substep satisfies detailed balance and the first law with zero external work. The positive actuation heat comes from a decrease in mean interaction energy. The example accounts for relaxation of an initially nonequilibrium law; a maintained supply of useful work or a learning process would require additional dynamics and resources.

`Examples/AgencyCycle.lean` specializes the general composition, total heat budget and first law to these same channels and laws. A biased initial memory law gives an asymmetric intermediate law and a sensing step that changes the joint distribution, with positive actuation heat. The regression `mismatched_intermediate_rejected` excludes a second step initialized without the required coordinate swap; the symmetric equilibrium law alone cannot detect that error. The reciprocal cycle also rejects a zero total heat budget.

Finite goal-directed control. `FiniteControlProblem`, in the same module, supplies a common initial law p_0 , world channel $K_a(t | s)$, memory channel $U_t(y | x)$ and terminal reward $r(y, t)$. A deterministic policy $a = \pi(x)$ induces the complete update above with $K_x = K_{\pi(x)}$. The theorem `cycle_law` identifies its final law with the channels carrying this explicit policy. Action remains a function of existing memory, and observation is the world state. No stochastic action paths are discarded before calculating heat.

The declared reservoir model assigns $q_A^\pi(x, s, t) = \theta \log[K_{\pi(x)}(t | s)/K_{\pi(x)}(s | t)]$ and $q_M(t, x, y) = \theta \log[U_t(y | x)/U_t(x | y)]$. Their thermal interpretation is the local-detailed-balance modelling input. The performance $J(\pi) = \sum_{y,t} p_2^\pi(y, t) r(y, t)$ and total mean heat $Q(\pi) = \langle Q_A^\pi \rangle + \langle Q_M \rangle$ use the same process, including its policy-dependent intermediate law. For a finite permitted set Π with at least one feasible member, `exists_optimal` proves

$$\exists \pi_* \in \Pi : \quad Q(\pi_*) \leq B, \quad J(\pi) \leq J(\pi_*) \quad \text{for every } \pi \in \Pi \text{ with } Q(\pi) \leq B. \quad (6)$$

Neither strict improvement nor a task objective follows from existence alone.

`Examples/AgencyControl.lean` supplies a lamp task with all four deterministic bit policies permitted. Initially X is uniform and independent of a lamp S with $\Pr(S = 1) = 1/4$. The fixed world channel sets $S' = a$ with probability $3/4$; memory then sets $X' = S'$ with probability $3/4$. The reward is $r(y, t) = \mathbf{1}_{t=1}$, and the task target is $J \geq 1/2$. At $\theta = 1$, use the same energy $E(x, s) = (\log 3)\mathbf{1}_{x \neq s}$ for every policy, with transition work defined by the two first laws above. Lean proves the following exact expectations from the finite path laws:

Policy	J	$\langle Q_A \rangle / \log 3$	$\langle Q_M \rangle / \log 3$	$Q / \log 3$	$\langle W \rangle / \log 3$
$\pi(x) = 0$ (baseline)	$1/4$	0	$1/4$	$1/4$	0
$\pi(x) = x$	$1/2$	$1/4$	0	$1/4$	0
$\pi(x) = 1 - x$	$1/2$	$1/4$	$1/2$	$3/4$	$1/2$
$\pi(x) = 1$	$3/4$	$1/2$	$1/4$	$3/4$	$1/2$

All policies start at mean energy $(\log 3)/2$ and end at $(\log 3)/4$; `first_law` gives $\langle W \rangle = Q - (\log 3)/4$. Work can be nonzero on individual paths even when its expectation is zero. At $B = (\log 3)/4$, `unique_optimum` identifies copying as the unique feasible maximizer. It doubles performance over the feasible baseline and attains the task target. The baseline misses that target despite having the same heat and mean work. The constant-true policy performs better but is infeasible. Copying and inversion have identical final joint laws and unequal heat costs, so even the full endpoint law cannot replace path accounting. Both selected substep channels allow state changes, and the selected policy satisfies the complete-update entropy bound.

The reward, permitted policies, initial resource and budget are declared model inputs. Selection compares fixed autonomous policies for one update; it accounts for neither installing or switching them nor learning their values. The result supplies no biological identification, continuing energy source, register-to-actuator allocation or spatial coupling convergence.

Learning dynamics of a policy register. A specified learning model assigns a physical bit b to the two feasible lamp policies: π_0 is the baseline and π_1 copies memory. At thermal scale one, choose register energy

$$E_L(b) = 4 \log 3 [J(\pi_1) - J(\pi_b)], \quad T = \begin{pmatrix} 5/8 & 3/8 \\ 1/8 & 7/8 \end{pmatrix}, \quad (7)$$

with rows and columns ordered $0, 1$. Thus $E_L(0) = \log 3$ and $E_L(1) = 0$. The energy encodes supplied expected task values; no observations or reward estimator are part of this model. Every transition has

outward heat $q_L(b, c) = E_L(b) - E_L(c) = \log[T_{bc}/T_{cb}]$ and zero work. This is a declared autonomous reservoir model whose detailed-balance equality is checked on all paths.

The construction `FiniteFeedbackStep.iterate` initializes each update from the preceding final law. Its `sum_entropy_balance` and `sum_first_law` telescope entropy and energy over any finite number of updates, preserving strict support. In `Examples/PolicyLearning.lean`, $p_0 = \Pr(b_0 = 1) = 1/4$ gives

$$p_{n+1} = \frac{3}{8} + \frac{1}{2}p_n, \quad p_n = \frac{3}{4} - \frac{1}{2} \left(\frac{1}{2}\right)^n, \quad (8)$$

$$J_n = (1 - p_n)J(\pi_0) + p_nJ(\pi_1) = \frac{7}{16} - \frac{1}{8} \left(\frac{1}{2}\right)^n. \quad (9)$$

The theorem `performance_improves` proves strict expected improvement at every finite update. `law_limit` and `stationary_law` identify convergence to the stationary law with $\Pr(b = 1) = 3/4$; `performance_limit` gives $J_n \rightarrow 7/16 < 1/2 = J(\pi_1)$. These are exact finite-state results, not simulation estimates.

Heat is evaluated on the same evolving forward path laws:

$$\langle Q_n^L \rangle = \frac{\log 3}{4} \left(\frac{1}{2}\right)^n, \quad \sum_{n=0}^{N-1} \langle Q_n^L \rangle = \frac{\log 3}{2} \left[1 - \left(\frac{1}{2}\right)^N\right] = \langle E_L \rangle_0 - \langle E_L \rangle_N. \quad (10)$$

The cumulative heat is bounded by $(\log 3)/2$ and satisfies the summed KL-based entropy balance. The energy comes from the prepared register's relaxation; this finite total supplies no continuing power source.

Both decoded policies have lamp task heat $(\log 3)/4$ and zero mean task work, so their mixture retains those prospective expectations for every n . This evaluation presumes the lamp task's prepared initial law, independent of the policy register. The learning channel executes no task episodes, resets, reward evaluations or actuator installation, and its heat does not pay for those operations. In particular, summable register heat does not bound the cost of repeatedly performing the task.

The regression `reward_decreasing_path_positive` retains positive probability of a copy-to-baseline transition. The same channel initialized with $\Pr(b = 1) = 7/8$ lowers expected reward (`overprepared_reward_decreases`). Improvement therefore depends on the objective, channel and initial preparation. This model establishes policy adaptation toward encoded task values. The model below supplies no such values; cortical identification and the spatial coupling convergence of E45Active require further models.

Learning an unknown task from observed outcomes. Which of two actions is rewarded is an environmental bit w that the learner never reads. Its register bit r is read out as the executed action $a = r$; the world returns success with $\Pr(o = 1 \mid w, a) = 3/4$ when $a = w$ and $1/4$ otherwise; and the update is win-stay, lose-resample, so $o = 1$ leaves r unchanged and $o = 0$ redraws it uniformly. The fences are signatures rather than prose. In `FiniteObservationalLearner`, the update has type $O \rightarrow R \rightarrow \text{ProbDist } R$ and the readout has type $R \rightarrow A$: neither takes the parameter as an argument, so no channel can read it and no precomputed policy value can be decoded into the action. The reward enters `performance` and no channel.

Acting, observing and updating compose into one `FiniteFeedbackStep` on (W, R) with the parameter held fixed, so the existing path-law entropy balance, local detailed balance, first law and their telescoped sums apply to exactly the process the reward is read from. Its composite channel is

$$T_w = \begin{pmatrix} 5/8 & 3/8 \\ 1/8 & 7/8 \end{pmatrix} \quad (w = 1), \quad (11)$$

with rows and columns ordered $r = 0, 1$, and its mirror image at $w = 0$; every entry is positive. From the uniform prior on (w, r) , `law_apply` gives every mass of the actual joint law at every horizon, and hence the expected reward of the action actually executed:

$$\Pr_n(w, r) = \frac{3}{8} - \frac{1}{8} \left(\frac{1}{2}\right)^n \quad (r = w), \quad \Pr_n(w, r) = \frac{1}{8} + \frac{1}{8} \left(\frac{1}{2}\right)^n \quad (r \neq w), \quad (12)$$

$$J_n = \frac{3}{4} - \frac{1}{4} \left(\frac{1}{2}\right)^n. \quad (13)$$

`performance_improves` proves strict improvement at every update from $J_0 = 1/2$, `performance_limit` gives $J_n \rightarrow 3/4$ and `performance_below_certainty` keeps every J_n below 1. `action_marginal`

unchanged shows that each action is executed half the time at every horizon: what the learner acquires is a correlation with the parameter, not a preference between actions.

Two baselines share the task, prior, readout and resource allowance. **frozen_performance** holds the register fixed and **blind_performance** replaces the world by one returning success with probability 1/2 whatever is done; both stay at 1/2 for every n , and **blind_heat** shows the second spends nothing while remaining strictly positive. **improves_on_baselines** states the strict improvement over both within the common allowance. The claim therefore rests on the observations carrying task information, not on the update rule or on the heat. **observation_informative** exhibits that information as conditional on the action, $\Pr(o = 1 \mid w = 1, r = 1) = 3/4$ against $1/4$ at $w = 0$, while **outcome_marginal_uninformative** shows the outcome's marginal carries nothing about w at the prior.

Heat is the transition log-ratio of the same composite channel at thermal scale one, $q_w(r, r') = \pm \log 3$ on the two register flips and 0 otherwise, and is evaluated on the actual evolving laws:

$$\langle Q_n \rangle = \frac{\log 3}{8} \left(\frac{1}{2} \right)^n, \quad \sum_{n=0}^{N-1} \langle Q_n \rangle = \frac{\log 3}{4} \left[1 - \left(\frac{1}{2} \right)^N \right] \leq \frac{\log 3}{4}. \quad (14)$$

The register's two states are energetically degenerate, so it stores nothing and **cumulative_work** makes that whole sum external work. The work is irreducible rather than a modelling convenience. **zero_work_needs_parameter_independent_heat** proves that a register energy giving zero work on every path forces $q_w = q_{w'}$ for all parameter values, and **no_zero_work_energy** exhibits $q_1(0, 1) = \log 3 = -q_0(0, 1)$: an energy landscape that paid for this learning would be one that already encoded the answer. **learning_entropy_budget** closes the accounting by telescoping the KL-based entropy balance over the same laws, so the joint entropy the learner removes fits inside $(\log 3)/4$.

reward_decreasing_transition retains probability 1/8 that the register leaves the rewarding action, so expected improvement is not pathwise improvement, and **successive_laws_differ** rejects a frozen register and a reset to the prior. Preparing the prior, supplying the work that drives the updates, executing task episodes beyond the declared updates, fabricating the readout, convergence to an optimal policy and any cortical identification are outside this model.

A continuing agent with a finite work store. **FiniteProtocol** in **Phase3.ContinuingAgent.lean** allows a different channel at every stage, carrying each actual final law into the next initial law. **sum_entropy_balance**, **sum_first_law** and **cumulative_entropy_budget** telescope the existing finite path balances over this sequence without stationarity. **ContinuingProcess** adds a common energy, stage heat observables, a thermal scale and an initial work allowance B . Its predicate **Sustains N** requires $B - W_{<k} \geq 0$ for every $k \leq N$, where work is an expectation on the executed paths. The resulting bound is the article's Eq. (11); when each stage costs at least $c > 0$, **horizon_le_of_cost** gives $Nc \leq B$. These results include the system's mean energy drop and do not identify the allowance with a register's erasure heat.

ContinuingAgent implements a three-stage cycle on (w, r, e) : the parameter w is fixed, the register r selects the action, and the environment flag e retains what that action did. Acting drives the flag while the register drifts; learning reads the flag, writes the register and lets the flag drift; reset prepares the flag for reuse while the register drifts. The learner's readout, observation and update do not take w as an argument. The actuator responds to w , as a physical task must, and reward is used only to score the law. The observation is marginalized inside the learning channel, whose thermal interpretation is supplied by local detailed balance for that composite channel.

In **Examples/ContinuingAgent.lean**, all five state types are bits and the initial joint law is uniform. A clear flag is set with probability 3/4 by the rewarding action $r = w$ and 1/4 by the other; an already set flag stays set with probability 3/4. Observation is correct with probability 3/4. Success keeps the register with probability 3/4 and failure flips it with that probability. Idle drift keeps a bit with probability 3/4; reset sets the flag's probability to 1/4 independently of its input. The three composite channels have strictly positive support. With common energy zero, thermal scale one and stage heat equal to each channel's forward/reverse transition log-ratio, every path's work equals its heat.

The proved mass recursion gives agreement $a_0 = 65/128$ after the first cycle and

$$a_{m+1} = \frac{65}{128} + \frac{a_m}{128}, \quad J_{3(m+1)} = a_m \geq \frac{65}{128} > \frac{1}{2}. \quad (15)$$

The flag has probability 1/4 and is independent of parameter and register at each completed reset. Successive cycle laws differ. These are exact finite-channel identities, proved in Lean rather than fitted

to a simulation. On the same laws, `first_cycle_work` and `totalWork_cycles_lower` give

$$W_{<3} = \frac{15}{32} \log 3 + \frac{1}{16} \log \frac{5}{3}, \quad W_{<3m} \geq m \left(\frac{3}{16} \log 3 + \frac{1}{16} \log \frac{5}{3} \right). \quad (16)$$

The learning stage costs $(\log(5/3))/16$ at every cycle, and reset has strictly positive mean heat. Every prefix of the first cycle fits the declared $B = 4 \log 3$, while 22 complete cycles exceed it. This is a conservative exclusion, not the exact exhaustion time.

Two controls delimit the result. Uniform observations leave chance reward at every completed cycle and zero learning-stage heat. Replacing only reset by idle drift leaves flag probability $17/32$ after the first cycle, against $1/4$ with reset, and reduces second-cycle reward from $8385/16384$ to $4157/8192$. The control still learns; reset restores the specified workspace law and changes subsequent behaviour. The store limits expected work at each prefix, not the battery on every realization. Initial-law preparation, the initial store, microscopic battery and sensor-memory dynamics, fabrication, optimal-policy convergence and cortical identification remain supplied or outside the construction. Its indefinite law sequence is a mathematical continuation, whose funded prefixes are selected by `Sustains`.

Active and passive composition (Chain.lean). The theorem `chain` uses the passive predictive node; `chain_active` uses the article's Eq. (12) for a named `FiniteFeedbackStep` M , heat observable q , thermal scale θ and budget B . The process's controller has the register's state type. `E34Active` requires positivity, local detailed balance at that register's temperature and $\langle Q \rangle_M \leq B$, with B the register's heat. The entropy lower bound is derived by `activeBound_of_e34Active`. A lower bound on erasure heat supplies no upper bound for an actuator and equates no operations; the upper-budget allocation comes from a resource model instead. `RegisterLedger` carries one register's update, its operations as elementary steps controlled by that register, local detailed balance for each at that register's own temperature, and the accounting identity $Q_{\text{reg}} = \sum_i \langle Q \rangle_i$ for the heat its operations deliver to its reservoir. `opHeat_nonneg_of_compressive` applies the left inequality of the article's Eq. (12) to an operation with $H(p') \leq H(p)$, so its share is nonnegative; `opHeat_le_dissipation` then bounds an operation whose companions all compress by Q_{reg} , and `e34Active_of_ledger` turns that bound into the edge. Under Landauer's heat equation, `ledger_bathEntropy` exhibits the same total as the register's own bath entropy change. The identity and the compressiveness are the model's physical inputs. Equation (4) accounts for both finite update substeps at a common thermal scale; it supplies neither this allocation nor the coupling-convergence implication of `E45Active`, and the ledger's index labels operations rather than sequencing them into one evolving law.

`E45Active` consumes `ActiveBound` for these same M, q, θ, B and requires convergence of the specified coupling-energy sequence to its named limit. Neither an unspecified agent nor a bound for a different process discharges this edge. The scalar constitutive model below discharges this edge for its own process-dependent density, and the microscopic model below discharges it for a kernel on the product space built from mode occupancies the step actually changes. `E45` is discharged the same way for named hardware holding a declared `PredictiveDissipation`'s own joint law. Both proofs discard their premise, because the refinement data and not the bound supplies convergence. Both branches then use `chain_from_coarseGrains`, sharing `E56–E89` and preserving the particular cover and its fixed section. Each composition has eight named edges; local-state compatibility, contraction and the fixed-state identification remain supplied physical content.

`chain_active_hypotheses_jointly_satisfiable` combines the noisy actuator with the one-bit register at thermal scale one, the changing grid and the specified field/cover. Its heat fits within the register's $\log 2$ budget, while its passive signed information difference is negative. `thermalAgency_zero_budget_rejected` excludes a zero budget for that same heat observable. `thermalAgency_wrong_limit_rejected` exhibits a valid active bound for which the constant-zero energy sequence cannot converge to the named limit three. The positive mesh witness supplies convergence independently; joint satisfiability establishes no common mechanism or thermodynamic derivation of convergence.

The register's ledger (Examples/RegisterBudget.lean). The witness is that same one-bit register, at temperature one with Landauer heat $\log 2$, and its operations act on a lamp bit through one family: the environment relaxes toward agreement with the held register bit at odds $a : 1 - a$, from a law whose agreement probability is w , with interaction energy $\log[a/(1-a)]$ for disagreement and mean heat $\log[a/(1-a)](a-w)$. The controlled operation takes $a = 2/3$, $w = 4/15$ and delivers $(2/5) \log 2$; the other takes $a = 4/5$, $w = 1/2$ and delivers $(3/5) \log 2$. Their sum is the register's $\log 2$, so `registerBudget_e34Active` discharges the bridge with no numerical comparison between separate models, and `chain_active_budget_jointly_satisfiable` composes the whole branch through it. Both inputs are load-bearing. `act_not_compressive` shows the controlled operation increases the joint entropy, so its own

second law bounds it neither below at zero nor above; `compressive_exceeds_budget` exhibits an entropy-preserving operation of the same family delivering $(7/3) \log 2$, above the register's whole dissipation; and `leaky_exceeds_budget` exhibits a second ledger for the same register, satisfying positivity, local detailed balance and the identity, whose other operation draws heat out of the reservoir and whose controlled operation is allocated $(7/6) \log 2$. `leaky_other_not_compressive` identifies that operation as the one the derivation excludes.

The register's bath (Examples/RegisterBath.lean). `FiniteProtocol.energyTransfer` is the gain of the subsystem whose energy is declared, on an actual transition. `sum.energyTransfer` and `reported_heat_eq` (`Phase3.ContinuingAgent.lean`) give the article's Eq. (16) for any finite protocol: the first is a first-law statement that admits deterministic gates and zero masses and asserts no entropy or temperature identification, and the second exhibits the correction by which a reported ledger differs from the modelled bath's gain.

The witness carries one register bit and one bath bit. `lift` sends each of the four maps of the register to a permutation of the pair (`lift_injective`) that performs the requested map on a bath prepared `false` (`lift_realizes`); an injective map acts on the register alone and an erasure swaps the register's bit into the bath. The prepared law is a uniform register and a pure bath, the bath's levels are 0 and $2 \log 2$, the register's are degenerate (`bathEnergy_register_degenerate`), and the thermal scale is one. `operationHeat` is the bath's mean energy gain on the executed paths and `operationHeat_formula` evaluates it as $\log 2$ for the two erasures and 0 for the two injective maps. A `StatisticalMechanics.Bool` instance discharges Landauer's heat equation for these values, with the reachable bath supports as its final supports; the instance is local to the file, so §1's bath, which charges $\log 2$ for every map including the identity, and its consumers are unchanged.

Three results fence a map-indexed budget. *The eraser is not reusable.* `erase_law_one` clears the register and leaves the bath uniform, and `erase_law_two` proves that a second execution of the same gate restores the prepared law, so the register is uniform again. The shares are $\log 2$ (`erase_first_heat`) and $-\log 2$ (`reused_bath_heat`) about a total of zero (`two_stage_heat`), which is the generic bath ledger applied to this protocol rather than a separate calculation, so a share is neither nonnegative nor bounded by the total (`shares_exceed_total`). *The register's map does not fix the cost.* `driveGate` holds the register at its value and excites its bath, so it performs the identity on a prepared bath at heat $2 \log 2$ rather than 0 (`drivenIdleHeat_eq`, `same_update_different_heat`). The work is accounted with the bath: $\log 2$ for one erasure (`erase_work`), $2 \log 2$ for the drive (`driven_work`) and zero for two erasures (`reuse_work`), the register's levels being degenerate. *Reversibility supplies neither ledger premise.* A deterministic gate sends one state to one state, so its channel charges nothing to any other (`gate_channel_zero`) and the reduced step is not strictly positive (`gate_not_positive`); and a swap is an involution, so its forward and reverse transition masses agree on every realized path and its log-ratio heat vanishes at every thermal scale (`erase_stageHeat_zero`) while its bath gains $\log 2$ (`erase_stageHeat_not_bath_heat`), the whole of which the reported ledger then omits (`erase_reported_correction`).

This constructs no `RegisterLedger` instance and weakens none of its results: positivity, local detailed balance and the accounting identity remain the physical inputs `RegisterBudget` supplies, and this model shows that a reversible microscopic gate does not supply them. The values follow from the declared energy gap and preparation. Preparing the bath, supplying the work, fabricating the gate and identifying this bath with the reservoir the ledger's operations exchange heat with remain separate.

Actuated scalar coupling (Phase3.ActuatedCoupling.lean). `ActuatedCoupling` carries a named feedback step, temperature and heat, a nonnegative conversion gain g , a spatial profile b , and regular meshes of a finite-measure region whose cell diameters tend to zero. Its signed drive $d = H(p) - H(p')$ is computed from the step's actual initial and final joint laws. The constitutive choice $f_A(x) = g db(x)$ fixes the density consumed by `discreteEnergy`; it is not an independently supplied energy sequence. Uniform continuity and integrability of b give the article's Eq. (13): `actuated_coarseGrains` applies mesh refinement to this density, and `e45Active_of_actuatedCoupling` discharges the corresponding edge. The budget plays no role in convergence. `actuated_limit_le_budget` uses that same process's `ActiveBound` to bound the limit, provided the total profile is nonnegative.

The noisy two-bit actuator has positive drive $d = (3/4) \log 3 - \log 2$. The witness in `Examples/ActuatedCoupling.lean` uses the nonconstant profile $b(x) = 1 + |x - 1/2|$ on $[0, 1]$. At unit gain the first two spatial energies are $3d/2$ and $5d/4$, so refinement changes the energy. The stationary version of the same channel has zero drive and actuates zero density. Doubling the gain strictly increases the limit at unchanged heat, and replacing the profile by one changes the spatial energy. Calibrating g to give limit three lets `chain_active_actuated_jointly_satisfiable` compose with the existing field and cover witnesses; a zero target is rejected for that same energy sequence. The example retains

the thermal actuator’s numerical budget comparison, separately from the register-ledger witness. All numerical identities here are exact Lean statements.

The constitutive gain and profile are physical inputs. The drive is an ensemble entropy difference, and its identification with a spatial amplitude is the supplied mechanism law. This model neither derives that law from local actions nor accounts for the work of installing it. The spatial mesh index is independent of learning time. The result is a scalar density and energy limit; the product-space kernel and cortical identification remain separate inputs.

Microscopic actuation (*Phase3_LocalActuator.lean*). `LocalActuator` carries mode occupancies c_i on configurations, fixed spatial profiles ϕ_i and installation prices u_i ; its `kernel` and `storedEnergy` are the two sums of the article’s Eq. (14). `kernel_update` computes a transition’s kernel change from its changed occupancies alone, and `kernel_update_local` leaves a pair untouched when every changed mode has zero profile at one of its two points; `kernel_symmetric` and `kernel_nonneg` record reciprocity and, for nonnegative occupancies and profiles, positivity. `pathWork` is $U(z') - U(z) + q$ along one path, and `installation_first_law` evaluates its mean on the executed step as the change in expected stored energy plus that step’s mean heat. `expected_kernel_update` states the ensemble kernel change as the mean of those same pathwise changes, so an unrelated final law or a prospective policy evaluation does not substitute for the executed one.

`KernelMesh` (*Phase2_KernelMesh.lean*) is a finite measurable partition with one sample per cell, whose reconstruction extends each sampled weight throughout the whole cell pair. `energy_eq_integral` proves the cell-mass weighted finite energy to be exactly that reconstruction’s product-measure integral; it needs finite mass and measurable cells, and neither continuity nor refinement. `kernel_tendsto` and `energy_tendsto` then give the article’s Eq. (15) for a continuous kernel on a compact second-countable substrate of finite mass, with `snap_tendsto_of_dist` reducing the sample hypothesis to a shrinking distance bound. `KernelArrangement` bundles the hardware, its configuration law, the substrate measure and the partition sequence, and its `coarseGrains` is the limit. In *Chain.lean*, `e45Active_of_localActuator` discharges the active edge for an arrangement holding a named step’s final law, and `e45_of_localActuator` discharges the passive edge for one holding a declared predictive system’s joint law.

The witness in *Examples/MicroscopicCoupling.lean* is one response mode on $[0, 1]$ with profile $(4/3)(1 + x)$ and unit price, held by the two-bit thermal channel: the mode is occupied exactly when controller and switch agree. Before the channel the installed kernel is $(8/9)(1 + x)(1 + y)$ and after it $(4/3)(1 + x)(1 + y)$, so executing the channel strictly raises a pairwise value; closing the switch takes the pathwise kernel at the origin from 0 to $16/9$. The mean external work is $(1 + \log 3)/4 > 0$, which the first law splits into the unit stored energy at probability one half and the channel’s $(\log 3)/4$ of heat, and a zero-work claim is rejected for those same paths. Clamped left-endpoint partitions into $n + 1$ cells have sample error at most $1/(n + 1)$; the first two coupling energies are $4/3$ and $25/12$, and the limit is 3. That limit matches the existing field witness, so `chain_active_microscopic_jointly_satisfiable` composes the active branch through a kernel on the product space, and a zero target is rejected for the same sequence. The unit interval is atomless, so these energies are cell masses rather than vertex weights. Reading §18’s predictive law on the same hardware installs $(16/9)(1 + x)(1 + y)$ with limit 4, so the configuration law is load-bearing, and §18’s two systems differ by their whole dissipated work while discharging the identical passive edge.

Mode profiles, prices, occupancy readout, the reservoir convention and the substrate measure are declared hardware data. The model accounts for one prepared update: fabricating the hardware, preparing its initial configuration law and supplying continuing power are separate. The spatial index is independent of the number of executed updates, and convergence in one is not convergence in the other. Nothing here identifies the substrate with cortex or the modes with a physical field, and no argument makes predictive efficiency select this hardware.

The class field in *Phase3_KLBound.lean* is separate from Gibbs’ nonnegativity theorem. The witness `boolDetuned` attains the assumed bound with $D_{KL} = \sigma = \log 2$, showing that a positive divergence is permitted.

The fixed-phase squared-drift functional in *Phase8_ContinuousField.lean* is a quadratic functional of an affine kernel map. The bounded operators `kernelCLM` and `continuumDriftCLM`, together with the Hilbert-space adjoint, give $\nabla\sigma = (2/D)A^*(AK + \omega)$. The theorem `structural_resonance_decreases_sigmaContinuum` proves descent for the specified gradient flow, with an atomless witness on the unit interval. The identifier `entropy_production_rate` names this squared-drift functional in the source; the name does not establish its identification with thermodynamic entropy production. That interpretation requires probability-current information absent from this calculation.

For a symmetric kernel, antisymmetry of $K(x, y) \sin(\theta_y - \theta_x)$ makes its double integral zero. Thus the mean drift is independent of the phase field. A dynamically phase-locked field attains the associated squared-drift lower bound under the stated hypotheses. The theorem `phase_locked_minimizes_entropy_of_symm` expresses this minimization. A one-way two-site coupling supplies a counterexample without symmetry. These are statements about the objective and its constrained mean, not convergence of joint noisy phase/plasticity dynamics.

4 Macroscopic Scaling and Kuramoto Phase Synchronization

We posit a continuous macroscopic coupling description. We model the local phases θ_i as coupled non-linear oscillators via the Kuramoto model.

Scope of the continuum description. `mesh_refinement_convergence` proves convergence of one scalar coupling energy. It produces neither a continuum kernel nor field dynamics: `edge_region` has no $M \times M$ product structure from which such a kernel could be recovered, and the vertex-supported candidate is invisible to the continuum functional on an atomless substrate. Both obstructions are to this architecture: `KernelMesh` carries cell pairs of positive mass and its limit is a kernel integral, at the price of declaring the hardware that sets the weights. The symmetric kernel is therefore a modelling input either way. Its identification with the brain’s endogenous *electromagnetic* field is a further empirical commitment about cortex, argued in the main text from the ephaptic-coupling literature with explicit calibration and observational limitations. Nothing in this section mathematically establishes either step.

For a system with coupling strength K and thermodynamic noise D , the dynamic evolution is:

$$\dot{\theta}_i = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i) + \eta_i(t) \quad (17)$$

Theorem 4 (Stationary coherent branch): Under the identical-frequency von Mises stationary ansatz, the self-consistency equation has exactly one positive solution when $K > 2D$ and none when $K \leq 2D$. This does not assert dynamical selection or noiseless phase locking in the stochastic model.

Lean 4 Implementation (Phase4.KuramotoDynamics.lean): We formalize the Lyapunov potential $V(\theta) = -\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_j - \theta_i)$ of the coupled field and prove two directions: `phase_locked_minimizes_potential` (the phase-locked configuration attains the global minimum whenever all couplings are positive) and `potential_min_implies_phase_locked` (any minimizer is phase-locked). Note the scope: this is a statement about the *static* potential landscape and holds for all positive coupling matrices. The critical threshold $K_c = 2D$ does not appear in this file. It is formalized separately, in `Phase8.SelfConsistency.lean` (see below), as a statement about the mean-field self-consistency equation with the stationary density assumed; deriving that density from the noise-driven stochastic dynamics would require the spectral theory of the associated Fokker-Planck operator, which is not available in Mathlib. The numerical controls below distinguish identical-frequency diffusion from the separate quenched-frequency model illustrated by `simulations/kuramoto.py`.

Positivity, and the regime these theorems cover. The hypothesis “all couplings positive” is not a technical convenience and it is not free. `KuramotoSystem` carries $A : V \rightarrow V \rightarrow \mathbb{R}$, symmetric and otherwise unconstrained, so signed — frustrated — couplings are representable; but every result in this development that identifies *where* the dynamics ends assumes positivity. `phase_locked_minimizes_potential` and `potential_min_iff_phase_locked` assume $A_{ij} > 0$; the class field `ThermodynamicCover.A_pos` demands it of every instance, and so of every conclusion of Theorem 5; and `kuramoto_tendsto_global_minimum` requires the strictly stronger uniform bound $0 < a \leq A_{ij}$, since a enters the Łojasiewicz constant. Such a network is an unfrustrated ferromagnet with one minimum up to a global phase shift — exactly not the near-degenerate landscape a memory argument from geometric frustration requires. By contrast, nothing about the sign of A is used by `is_kuramoto_trajectory_exists` and its uniqueness counterpart, by the Lyapunov identity `dV_dt_le_zero` on which the whole descent argument rests, by `dynamic_potential_antitone` and `dynamic_potential_tendsto`, or by `velocity_sq_tendsto_zero`. Since positivity is a *field* of `ThermodynamicCover` rather than a hypothesis distributed across the development, it enters the chain at one node and not before it. So *that* the motion stops holds for frustrated systems as well; *where* it stops does not. On the continuum side `phase_locked_minimizes_entropy_of_symm` assumes only symmetry and so admits signed kernels, but it takes the existence of a dynamically phase-locked field as a hypothesis, which is precisely what frustration threatens. The main text scopes the frustration claim out of the chain accordingly, rather than repeating it beside theorems that exclude it.

Twisted gluing does not recover content from phase frustration (*Phase5-TwistedGluing.lean*). We also tested the natural weakening of exact overlap agreement: local states may differ by a phase on each overlap. `gluesUpToPhase_of_isCoboundary` proves that coboundary offsets glue after one rotation per patch, while `isCoboundary_of_gluesUpToPhase` proves the converse when the phase action is free on overlaps. Any Kuramoto configuration, including a twisted or splay state, supplies absolute patch phases ϕ_i , so its offsets $\phi_i - \phi_j$ are automatically a coboundary (`isCoboundary_of_phaseField`) and the twisted family glues (`gluesUpToPhase_of_phaseField`). The formal `PhaseObstruction` quotient is deliberately not called Čech cohomology because its total cochains still assign irrelevant values to patch pairs with empty overlap.

The data give a narrower positive answer. Townsend et al. (2015) identify phase singularities in multi-electrode primate LFP recordings by an integer winding number around electrode loops, and the associated complex waves modulate spiking; Xu et al. (2023) find task-dependent spiral centres and rotation directions in human fMRI phase maps. Around such a zero-amplitude point the circle-valued phase need not admit one real-valued lift. Local lifts differ on overlaps by integer multiples of 2π , supplying measured transition data without an arbitrary gauge field. *Phase5.PhaseLifts.lean* formalizes the finite electrode loop: `loopWinding` sums its integer transitions, `loopWinding_eq_zero_of_hasGlobalLift` proves that transitions from one global lift telescope to zero, and `threeElectrodeWinding_not_hasGlobalLift` gives a winding-one witness. This does not rescue `PhaseObstruction`: that quotient uses circle-valued differences of absolute patch phases, while the measured obstruction uses integer transitions between local lifts.

The cover-level specification has cortical surface M , analytic signal $z = Ae^{i\phi}$, singularity set Z_t , punctured domain $M_t^* = M \setminus Z_t$, and measurement map $q_t = z/|z| : M_t^* \rightarrow S^1$. A trial-independent electrode or source-space cover of M_t^* supplies local real lifts; their differences divided by 2π are integer transitions on connected overlaps, and nerve loops are the inputs to `loopWinding`. In finite-resolution data the amplitude threshold defining Z_t , the loop geometry and the required cross-scale persistence are fixed using training or baseline data, never the content labels of test trials.

The proposed content-sensitive test compares nested preregistered classifiers. The control model contains stimulus, task, behavioural response, eye movement and arousal; the augmented model adds signed winding, source-localized singularity position, rotation direction and lifetime. Evaluation uses held-out participants and sessions, with label permutations within task, stimulus and arousal strata. The content hypothesis requires replicated incremental out-of-sample discrimination, including under a changed or absent motor report. This controls the known task dependence rather than mistaking it for content. A positive result would be an empirical association, not an identification of a topological sector with phenomenal content; a null increment rejects this proposed role. The construction and test remain outside the conditional chain, and a full nerve-indexed integer complex is deferred until the measurement protocol warrants it.

Lean 4 Implementation (Phase8.ContinuousField.lean): The continuous-field file defines $K_c = 2D$ and a predicate for a substrate above that threshold. Two cautions. First, the predicate is not connected to `is_continuous_kuramoto_trajectory`, `entropy_production_rate`, or the order parameter of *Phase4*. What it is connected to is the mean-field self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$, whose bifurcation at $K = 2D$ we formalize in both directions (see the next paragraph): `exhibits_phase_transition_coherent` shows that a substrate satisfying the predicate admits a solution with $0 < r \leq 1$, and `exhibits_phase_transition_unique_coherent` that it admits exactly one. So the threshold is not a bare stipulation, but what is proved is a statement about a stationary density rather than about a trajectory. Second, the predicate requires the substrate's `volume` to be a probability measure, and this hypothesis is load-bearing rather than decorative. It compares `mean_field_coupling`, the kernel averaged over both arguments, against $2D$; on a substrate of total mass m that average carries a factor m^2 , so without normalization the comparison tracks the substrate's size rather than its coupling and can be satisfied by any kernel at all. *Examples §9* exhibits a system a factor of two below threshold whose unnormalized double integral nonetheless clears $2D$. Under the hypothesis, `mean_field_coupling_const` gives that a constant kernel has strength exactly K and `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$, witnessed in §9 on both sides of the threshold.

4.1 Stationary self-consistency

Lean 4 Implementation (Phase8.SelfConsistency.lean): The self-consistency equation behind $K_c = 2D$ is formalized as real analysis. With the von Mises stationary density $\rho(\theta) \propto e^{a \cos \theta}$, $a = Kr/D$, taken as given (the algebraic step from the stationary Fokker–Planck equation to this density is verified inde-

pendently at `simulations/fokker_planck_von_mises.py` in the repository), `besselRatio` is $I_1(a)/I_0(a)$ written as a ratio of interval integrals, Mathlib having no modified Bessel functions. `vonMisesM.eq_mul_vonMisesS` is the integration-by-parts identity $\int_{-\pi}^{\pi} \cos \theta e^{a \cos \theta} = a \int_{-\pi}^{\pi} \sin^2 \theta e^{a \cos \theta}$, exact because $\sin(\pm\pi) = 0$ kills the boundary term. `vonMisesC2.nonneg` proves $I_2(a) \geq 0$ by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ through the reflections $\theta \mapsto \pi - \theta$ and $\theta \mapsto \pi/2 - \theta$, where the integrand becomes a product of two manifestly non-negative factors; `vonMisesS.le_half_vonMisesZ` restates it as $\mathbb{E}_a[\sin^2 \theta] \leq 1/2$ and `besselRatio.le_half_self` as $R(a) \leq a/2$. That factor of $1/2$ is exactly what separates a threshold at D from one at $2D$; `subcritical_fixed_point_eq_zero` is the crude form obtained from $\sin^2 \leq 1$ and is kept to make the point visible.

The subcritical theorem is `subcritical_fixed_point_eq_zero'`: for $0 \leq K < \text{critical_coupling}$ D and $D > 0$, the only $r \geq 0$ with $r = R(K, r)$ is $r = 0$.

The supercritical theorem is `supercritical_fixed_point_exists`: for $K > \text{critical_coupling}$ D there is an r with $0 < r \leq 1$ and $r = R(K, r)$. Its proof needs no series expansion. Factoring $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the factorization the integration by parts supplies — it suffices that E is continuous with $E(0) = 1/2$: since $D/K < 1/2$, the map overshoots for small $r > 0$, while `besselRatio.le_one` (R is a mean of $\cos \theta$) makes it undershoot at $r = 1$, and `intermediate_value_Icc'` supplies the crossing. Continuity of E is Mathlib's continuity of a parametric interval integral, applied to the jointly continuous integrand. `critical_coupling_is_threshold` states the two halves together, and §9 of the file discharges the hypotheses of each at concrete values ($K = 3$, $D = 1$ above threshold; $K = 3$, $D = 2$ below it), so neither is vacuous.

Section 8 of the file identifies the equation with the physics it is named for. `vonMisesDensity` is the normalized weight, proved a probability density on $[-\pi, \pi]$ (`vonMisesDensity_integral_eq_one`); `circularOrderParameter` is the average of $e^{i\theta}$ against a density, the continuum analogue of `order_parameter_complex` of Phase4; `circularOrderParameter_vonMises` proves that average equals $R(a)$, real because `vonMises_mean_sin` gives $\mathbb{E}_a[\sin \theta] = 0$; and `fixedPoint_iff_selfReproducing` states the equivalence: r solves $r = R(K, r)$ exactly when the density it induces has order parameter r . `incoherent_density_uniform` identifies the solution $r = 0$ with the uniform density, and `supercritical_coherent_density` restates the existence theorem as the existence of a stationary density with positive order parameter.

Strict inequalities determine the threshold case. `vonMisesC2.pos` upgrades the fold argument to a strict inequality for $a > 0$ — `foldB` is continuous, non-negative on $[0, \pi/4]$ and strictly positive on its interior, so `intervalIntegral_pos_of_pos_on` applies — giving $\mathbb{E}_a[\sin^2 \theta] < 1/2$ and $R(a) < a/2$ for every $a > 0$. That closes the threshold case: `fixed_point_eq_zero_of_le_critical` covers all $K \leq 2D$, so the incoherent and coherent regimes exhaust $K \geq 0$ with no gap, and $K = 2D$ falls on the incoherent side.

`coherent_iff_sRatio_eq` reformulates the coherent branch: for $r \neq 0$, solving $r = R(K, r)$ is equivalent to $E(Kr/D) = D/K$, so the branch is a level set of E . `coherent_branch_continuous_at_threshold` then proves that for every $\varepsilon > 0$ there is a $\delta > 0$ such that on $(2D, 2D + \delta)$ every coherent solution satisfies $r < \varepsilon$ — the branch emerges continuously from zero, excluding a discontinuous jump. Its proof needs no monotonicity of E : on a compact interval bounded away from the origin the extreme value theorem gives a maximum strictly below $1/2$, and beyond that the crude bound $E(a) \leq 1/a$ suffices.

Strict decrease of E and strict increase of R give uniqueness and monotonicity of the coherent branch. Section 4.6 presents the folding and crossing argument. The consequences are `coherent_fixed_point_unique` (at most one positive solution), `supercritical_fixed_point_existsUnique` and `supercritical_solution_set` (the non-negative solutions are exactly $\{0\}$ at or below threshold and $\{0, r\}$ above it), and `coherent_branch_strictMono`, which uses both halves: E decreasing puts the concentrations of two couplings in the order $a_1 < a_2$ through $E(a_i) = D/K_i$, and R increasing carries that order to $r_1 < r_2$.

Scope: the density is assumed, not derived from the Fokker–Planck equation; nothing in this file differentiates anything, so E and R are reached as strictly monotone rather than as smooth, and the rate at which the branch leaves zero is the business of `Phase8CriticalExponent.lean` below; dynamical selection is not established, and could not be here, since no theorem in the file mentions a trajectory; and the order parameter reached is that of a density, not of a trajectory — no theorem relates `circularOrderParameter` to `order_parameter_complex`, which would be the mean-field limit. Every declaration in the file depends only on `propext`, `Classical.choice` and `Quot.sound`.

4.2 Stationary onset rate

Lean 4 Implementation (Phase8CriticalExponent.lean): The stationary onset rate follows from a second-order expansion of the integral ratio. `vmMoment n a` is the cosine moment $\int_{-\pi}^{\pi} \cos^n \theta e^{a \cos \theta}$, and `hasDerivAt_vmMoment` differentiates it in a under the integral sign, raising the cosine power by one; the interchange is Mathlib’s dominated-convergence form for interval integrals, dominated by the constant $e^{|a|+1}$ on the unit ball around the parameter, which is where the compactness of $[-\pi, \pi]$ and the boundedness of $\cos$ do their work. Iterating gives `contDiff_two_vmMoment`, and `vonMisesSRatio_eq_moments` rewrites $E(a) = 1 - \text{vmMoment } 2 a / \text{vmMoment } 0 a$, so E inherits two derivatives from a quotient whose denominator is positive.

The moments at $a = 0$ are elementary integrals of powers of $\cos$: 2π , 0 , π , 0 and $3\pi/4$. They give $E(0) = 1/2$, $E'(0) = 0$ — the first derivative vanishes by the symmetry that kills the odd moments — and $E''(0) = -1/8$, and Taylor’s theorem in Mathlib’s `taylor_isLittle0.univ` form then yields `vonMisesSRatio_second_order`: $E(a) - (\frac{1}{2} - a^2/16)$ is $o(a^2)$ at the origin. The remainder is $o(a^2)$ and not $O(a^4)$; the stronger form would need a fourth derivative and nothing here requires it.

`coherent_solution_critical_exponent` then combines the expansion with `coherent_iff_sRatio_eq`: writing $a = Kr/D$ and rearranging $E(a) = D/K$ gives $K - 2D = Ka^2(\frac{1}{8} - 2q(a))$ with q the normalized remainder, and letting $K \rightarrow 2D$ along any family of positive coherent solutions with $r \rightarrow 0$ sends $q \rightarrow 0$ and $r^2/(K - 2D) \rightarrow 1/D$. The statement is deliberately filter-indexed rather than parameterized by a curve, so it constrains every such family and privileges no path.

Because its hypotheses hold only eventually along the filter, and an eventual clause is vacuously true on the empty one, the file also exhibits a family satisfying it: `coherentBranch` selects the unique positive solution above threshold via `supercritical_fixed_point_existsUnique`, `coherentBranch_tendsto_zero` reads `coherent_branch_continuous_at_threshold` as a limit along $K \downarrow 2D$, and `coherentBranch_critical_exponent` discharges every hypothesis on it.

Scope is inherited unchanged from the file above: this is asymptotics of the solutions of a stationary equation. It establishes no stability, no dynamical selection and no time course, so reading it as a square-root foot on a recovery curve requires the adiabatic separation and a threshold crossing at non-zero speed, both of which the main text states as assumptions. Every declaration in the file depends only on `propxt`, `Classical.choice` and `Quot.sound`.

The gap between these stationary-density results and any statement about a finite trajectory is the dynamical mean-field limit — propagation of chaos for the Kuramoto system, which is a research programme rather than a lemma Sznitman (1991). No theorem in this development closes it.

4.3 Finite mean-field trajectory controls

Two finite- N controls probe this gap without closing it. First, 500 seeded replicas of 1,000 identical-frequency oscillators were integrated at fixed coupling. Their steady order was 0.06266 below threshold, 0.16049 at threshold and 0.67735 above threshold. The last value lies -0.00535 from the stationary coherent value 0.68270. Refining the integration step by a factor of 4 at the same coupling, duration and tail sampling moves that steady order to 0.68168, so most of the gap is Euler–Maruyama bias. The last refinement still moves it by 0.00147, so the remaining -0.00102 bounds the finite- N part rather than measuring it.

Early-growth fits are regressed over a window whose bounds follow the trace: it opens at $2/\sqrt{N} = 0.06325$, clear of the finite-size floor, and closes at half the static branch order, which runs from 0.20669 to 0.37050 across the swept couplings. The fits give $\lambda = 0.41505K - 0.78619$, against the infinite- N linearization $\lambda = 0.5K - 1$. That upper bound is set from the stationary branch rather than from the linear regime the fit assumes, so the finite window can itself contribute to the shortfall from $1/2$: the step refinement above is a control on stationary order and apportions no bias in this growth estimator. The sweep also integrates at the step whose stationary bias that control measures, and carries finite N ; this measurement separates none of the three. This exhibits finite sampled escape toward the coherent branch; it proves no stability or dynamical-selection theorem.

Second, 1,000 ensembles were swept from $N = 10$ to $N = 1000$ at $K = 1$ and $K = 3$. In the subcritical control, mean order fell from 0.3726 to 0.0402 and absolute circular pair correlation from 0.0458 to 0.0010. The matched-sample sliced joint-product distance changed only from 0.0468 to 0.0419, exposing its finite-sample floor. Above threshold, mean order stabilized at 0.7208, but the unconditional lab-frame pair correlation remained between 0.459 and 0.525 rather than following the $1/N$ reference. The unpinned collective orientation is a shared latent variable, so unconditional independence and independence conditional on that orientation are different claims. In the co-rotating frame, the $N = 1000$

density at measured concentration $a = 2.1624$ had L^1 error 0.1291 from the von Mises target and Bessel self-consistency residual 0.00191. These finite, single-seed diagnostics prove neither form of propagation of chaos and discharge no Lean obligation.

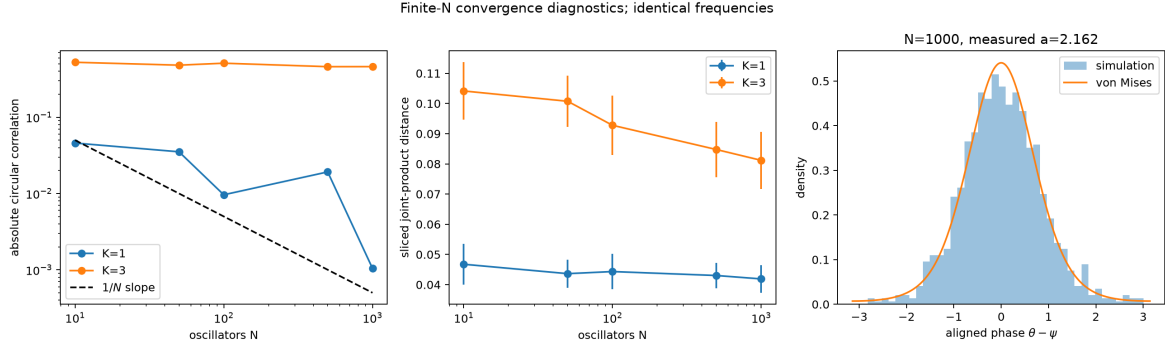


Figure S1: Finite- N convergence diagnostics for identical frequencies. The left panel compares branch-cut-invariant pair correlation with a $1/N$ reference; the centre uses matched sliced-Wasserstein joint-product samples; the right compares the co-rotating $N = 1000$ histogram with the von Mises density at measured concentration. The persistent supercritical lab-frame dependence is a negative result for the unconditional protocol, not a theorem about the conditional mean-field limit.

Dale-balanced rescue and calibration limits

Finite simulations separate synaptic balance from the magnitude of fluctuations. The network has no self-edges, an Erdős-Rényi edge probability of 0.2, and excitatory/inhibitory presynaptic fractions of 80/20. Each row's positive and negative weights sum to g and $-g$, respectively. With identical frequencies, $D = 1$, $N = 500$, $g = 1$, step size 0.01 and duration 100, three seeded runs give zero-field mean order in $[0.0508, 0.0574]$, close to the $1/\sqrt{N}$ reference. Controls at $g \in \{0, 1, 2, 4\}$ and $N \in \{250, 500, 1000\}$ show why row balance alone is insufficient to prescribe that reference: signed coupling can amplify finite-size fluctuations. Order decreases with size in both the weak and strong controls while $N\langle r^2 \rangle$ remains above the independent-phase value of unity; these finite sizes do not establish an asymptotic scaling law. The $g = 1$ regime is also below the positive-only mean-field threshold, so its low order does not establish a causal role for inhibition in preventing coherence.

A uniform per-pair term ϵ adds mean-field strength $K_{\text{eff}} = N\epsilon$. The operational crossing, defined by second-half mean $r = 0.2$ with all larger sampled couplings above it, occurs at $K_{\text{eff}}/D \in [1.96, 1.99]$ across the three seeds (Fig. S2). Each value interpolates across one step of the ϵ grid. That grid is geometric with ratio 1.136, and the one step of it the crossing falls in, $[1.857, 2.110]$, is subdivided linearly into intervals of width 0.025 and rerun on the same three seeds, which places the crossing in $[1.958, 2.009]$ — the union of the refined intervals the three seeds cross in. That bracket contains $K_c/D = 2$, so refining the grid tenfold locates the crossing without separating it from the mean-field threshold at $N = 500$. This range describes finite-run variability, not a confidence interval or an infinite-size critical point. Equal-duration half-step controls probe numerical sensitivity near the transition; they do not establish convergence of the integrator.

The field conversion requires an explicit calibration rate γ :

$$E_{\text{req}} = \frac{K_{\text{eff}}}{\gamma N_{\text{cortex}} \text{shift } f}.$$

Since shift f is dimensionless, γ supplies the inverse-time units of the dynamical coupling. Under the numerical convention $\gamma = 1$ in the simulation's inverse-time unit, the shared Fermi arithmetic gives field ranges $[0.5856, 0.5930]$, $[0.0732, 0.0741]$ and $[0.0217, 0.0220]$ mV/mm at decay lengths 0.1, 0.2 and 0.3 mm. These conditional numbers fall below the 1–5 mV/mm comparison band and scale inversely with γ . The simulation does not measure that calibration, so it cannot independently validate physical-unit consistency or a cortical field threshold. Neither rescue nor these controls discharge a Lean or gluing obligation.

No clinical mechanism for perceptual anomalies is inferred from the threshold calculation.

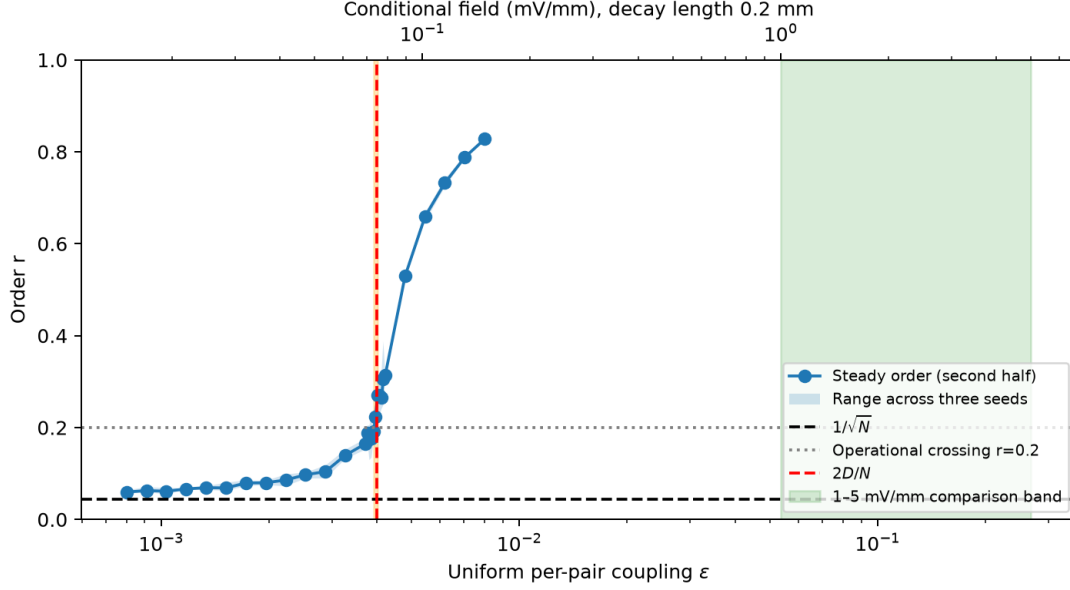


Figure S2: Finite- N rescue in the declared weak-synapse regime. Points are three-seed mean order; blue shading gives the seed range. Orange shading marks the union of sampled crossing brackets. The top axis uses the conditional $\gamma = 1$ field conversion at decay length 0.2 mm; green shading is the 1–5 mV/mm comparison band. The dashed horizontal line is $1/\sqrt{N}$. The balanced strong-synapse control remains a distinct regime and does not satisfy the floor-level baseline gate.

4.4 Deterministic trajectories and their limits

Lean 4 Implementation (Phase4_RotatingFrame.lean): The full potential includes the natural frequency term $-\sum_i \omega_i \theta_i$ that appears in the Lyapunov descent result `dV_dt.le.zero` in the combinatorial thermodynamics module. With any $\omega_i \neq 0$ the full potential is unbounded below (`kuramoto_potential_unbounded_below`), so it has no minimum for a phase-locked state to attain, so the reduced and full potentials must be distinguished. This file chains them by the standard rotating-frame reduction $\theta_i \mapsto \theta_i - \Omega t$. For a system of identical natural frequencies $\omega \equiv \Omega$: `is_kuramoto_trajectory_rotate` sends a trajectory to a trajectory of the zero-frequency system `sys.reduced` — the same system Theorem 5’s `ThermodynamicCover` works with; `kuramoto_potential_reduced` shows the two potentials coincide on it; `dynamic_potential_descent` transports the descent identity $\dot{V} = -\sum_i \dot{\theta}_i^2$ onto the potential above; and `rotating_frame_chain` runs this through the minimum characterization to phase-locking and $r^2 = 1$. Phase differences, phase-locking and the order-parameter magnitude are proved frame-invariant (`rotate_sub`, `is_phase_locked_rotate`, `order_parameter_r_sq_shift`), so no observable is lost. Examples §7 runs the chain end to end on a two-oscillator system, discharging every hypothesis, and exhibits a system whose full potential has no minimum. Scope: the reduction is exact only for identical frequencies. A spread of frequencies leaves residual detunings $\omega_i - \Omega$ in the reduced system, to which the unboundedness result still applies, and phase-locking then depends on K_c , which this file does not treat — see the `Phase8_SelfConsistency.lean` note above for what is proved about the threshold, and note that it concerns a stationary density rather than the trajectories this file transports.

Trajectories that run. The deterministic initial-value problem admits a unique global solution through each initial state. `kuramotoField_lipschitz` (`Phase3_CombinatorialThermodynamics.lean`) proves the Kuramoto vector field globally Lipschitz with constant $2\sum_{ij} |A_{ij}|$ — sin is 1-Lipschitz and the index set is finite, so there is no smallness caveat and the natural frequencies drop out. `is_kuramoto_trajectory_unique` turns this into Picard–Lindelöf uniqueness on all of $\mathbb{R}$: two solutions agreeing at one instant agree forever, in both time directions.

Existence follows by constructing solutions on arbitrary finite windows and gluing them using uniqueness. Mathlib’s `IsPicardLindelof` caps a local solution’s lifetime by the radius of the ball on which the field’s bounds hold, divided by the field’s sup norm there, and carries no global-in-time theorem; but `kuramotoField_norm.le` bounds this field by $\sum_i |\omega_i| + \sum_{ij} |A_{ij}|$ on the *whole* state space, so the cap can be defeated by enlarging the ball after choosing the window, and `kuramoto_exists_on_window`

produces a solution on every $(t_0 - T, t_0 + T)$. Uniqueness on an interval makes the family coherent — solutions on nested windows agree where both are defined — so $\theta(t) := \alpha_{[|t-t_0|]}(t)$ is a solution on all of $\mathbb{R}$ (`is_kuramoto_trajectory_exists`), and `kuramoto_cauchy_problem` states existence and uniqueness together as a single `ExistsUnique`: the trajectory predicate is inhabited exactly once for each initial state, rather than only by the solutions we write down by hand.

A two-oscillator solution illustrates the dynamics. On two oscillators with unit coupling and zero natural frequency the phase difference obeys $\dot{\Delta} = -2 \sin \Delta$, whose solution $\Delta(t) = 2 \arctan(c e^{-2t})$ splits symmetrically into an exact trajectory of the full system (`Examples §15`, `pairRelax_is_trajectory`). At $c = 1$ it starts a quarter turn out of phase (`pairRelax_not_locked_at_zero`); the cosine of the phase difference tends to 1, the order parameter tends to 1 through the two-oscillator identity $r^2 = (1 + \cos \Delta)/2$, and the dynamic potential tends to its global minimum. By the uniqueness result it is the only solution through its initial state.

Two general statements describe the long run, both for a zero-frequency system, which by `kuramoto_potential_unbounded_below` is the only case in which the potential is bounded below at all. `dynamic_potential_antitone` upgrades the instantaneous descent to monotonicity in t — `dynamic_potential_deriv_nonpos` constrains each instant separately and does not by itself say the trajectory settles — and with the uniform lower bound $-\frac{1}{2} \sum_{ij} |A_{ij}|$ monotone convergence gives `dynamic_potential_tendsto`: the potential converges along *every* trajectory.

Integrating the Lyapunov identity gives the quantitative form, `dissipation_integral_eq`: energy dissipated over $[0, T]$ is exactly the potential dropped, so by `dissipation_integral_tendsto` the improper integral $\int_0^\infty \sum_i \dot{\theta}_i^2 dt$ converges, to the total drop. The proof uses the derivative identity in `HasDerivAt` form, which includes differentiability and permits application of the fundamental theorem of calculus. `rotating_frame_dissipation` transports both to a uniform-frequency system.

Finiteness of that integral is not the same as $\sum_i \dot{\theta}_i^2 \rightarrow 0$: an integrable function can spike forever on ever narrower intervals. Barbalat’s lemma supplies the implication, proved here as `tendsto_zero_of_lipschitz_of_integral_le`: a non-negative Lipschitz function with $\int_0^T g$ bounded above tends to zero, because a spike of height ε drags a window of width $\varepsilon/2K$ up with it and only finitely many such windows fit under a finite integral.

Its hypothesis is then checked for the dissipation rate without differentiating it: the velocities are bounded by $\sum_i |\omega_i| + \sum_{ij} |A_{ij}|$, the field is Lipschitz in the configuration, and the configuration is Lipschitz in time because its derivative *is* the bounded field (`trajectory_dist_le`, by the mean value theorem, with no invariant region needed since the bound holds on the whole state space). The conclusion is `velocity_sq_tendsto_zero`: along *every* trajectory of a zero-frequency system, $\sum_i \dot{\theta}_i^2 \rightarrow 0$; equivalently each $\dot{\theta}_i \rightarrow 0$ and the vector field along the trajectory tends to 0 in the state space. The statement applies to arbitrary trajectories under these hypotheses, and `Examples §15` applies it to the relaxing pair, whose velocity starts at -1 and tends to zero.

At zero frequencies, the flow is the negative gradient of its dynamic potential. The identity $\dot{V} = -\sum_i \dot{\theta}_i^2$ is formalized by `dynamic_potential_hasDerivAt`. The minimizers of this potential are characterized by phase-locking. `lojasiewicz_estimate` adds the missing ingredient: on the region where every pairwise phase difference lies within a quarter turn, $2a^2W \leq \left(\sum_{ij} A_{ij}\right) \sum_i \dot{\theta}_i^2$, where W is the excess of the potential over its minimum and $a > 0$ is a lower bound for the coupling. Its proof uses no spectral gap and no convexity, only two extremal indices: at the site m carrying the largest phase every coupling term has $\theta_j - \theta_m \in [-D, 0]$ with $D \leq \pi/2$ the spread, so every term is non-positive, and the term from the site carrying the smallest phase is at most $-a \sin D$ (`velocity_max_le`); against $W \leq \frac{1}{2} \left(\sum_{ij} A_{ij}\right) (1 - \cos D)$ the estimate follows from $\sin^2 D = (1 - \cos D)(1 + \cos D)$ and $\cos D \geq 0$, which is the only place the quarter-turn hypothesis is used.

The initial energy bound keeps the trajectory in this region. Because all terms of the excess are non-negative, a single pair satisfies $\frac{1}{2}a(1 - \cos(\theta_j - \theta_i)) \leq W$, so an initial excess below $a/2$ — which `dynamic_potential_antitone` preserves — keeps every pairwise cosine strictly positive for all time. A difference escaping $[-\pi/2, \pi/2]$ would have to cross $\pi/2$ in absolute value by the intermediate value theorem, and there the cosine vanishes; so the quarter-turn region is forward invariant (`spread_le_small`). Continuity on the connected half-line thus pins the phase differences to that branch.

The differential inequality $\dot{W} \leq -cW$ with $c = 2a^2/\sum_{ij} A_{ij}$ then integrates, in the elementary form “ $W(t)e^{ct}$ has non-positive derivative”, to $W(t) \leq W(0)e^{-ct}$ (`excess_decay`). The rate transfers to the velocity through $|\sin d| \leq \sqrt{2(1 - \cos d)}$, giving $|\dot{\theta}_i(t)| \leq Be^{-ct/2}$ (`velocity_abs_le_exp`), which is integrable on $(0, \infty)$; since $\theta_i(t) = \theta_i(0) + \int_0^t \dot{\theta}_i$, the trajectory converges (`phase_tendsto`), and $W \rightarrow 0$ forces the limit to be phase-locked, hence a global minimiser with $r^2 = 1$ (`kuramoto_tendsto_global`).

minimum). `rotating_frame_tendsto_global_minimum` transports it to uniform frequencies, where what converges is the relative configuration.

The two hypotheses are on the initial data and neither can be removed: splay and twisted configurations are equilibria too, so “every trajectory reaches the phase-locked state” is false, and what is proved is the arc condition version — which the two-oscillator witness satisfies for every c , since $2 \arctan$ lands in $(-\pi, \pi)$.

Examples §17 runs the theorem on three sites, where the method of §15 is unavailable because the system has no closed-form solution: the trajectory is produced by `is_kuramoto_trajectory_exists` from the initial data $(0, 0, \frac{1}{2})$, which is not phase-locked, and its limit by the convergence theorem as $\theta(0) + \int_0^\infty \dot{\theta}$; the numerical content of checking the hypotheses is $\cos \frac{1}{2} > \frac{3}{4}$, supplied by `Real.cos_bound`.

4.5 Discrete coupling and mesh refinement

From continuous to discrete coupling: The coupling matrix A_{ij} above is not postulated. A triangulation of the substrate assigns to each edge (u, v) a region of the manifold, and the coupling weight is the Lebesgue–Bochner integral of the scalar magnitude of the stress-energy tensor over that region (`edge_weight`); symmetry and non-negativity of A are then proved rather than assumed. The bridge back is mesh refinement: as the regions shrink, the discrete energy $\frac{1}{2} \sum_{u,v} \mu(R_{uv}) f(x_u)$ should converge to the continuous energy $\int_S f$. Stated as a conjecture over the bare triangulation class, that is false: nothing there forces the regions to cover the manifold, empty regions have diameter zero, and the everywhere-empty triangulation refutes it. Stated correctly it is a theorem (`mesh_refinement_convergence`), quantified over a sequence of triangulations of a region S whose regions are measurable, pairwise disjoint across distinct edges, covering, and anchored to the embedded vertices, with mesh size tending to zero. The fixed-mesh error is bounded by $\varepsilon \mu(S)$, with ε a modulus of continuity of the integrand, and the symmetric double sum is proved equal to the midpoint rule over unordered edges.

Two limitations follow. The theorem gives convergence, not the $O(1/N^2)$ rate reported by `mesh_refinement.py`. And `TriangulatedManifold` itself is bare data: it pairs a complex, an embedding and a family of edge regions without relating them, so the geometric conditions live in a separate predicate `IsRegularTriangulation`. Because those conditions are in tension — cells disjoint yet covering, every vertex anchored to each of its edges — **Examples §6** exhibits an instance: the uniform partition of $[0, 1)$ into N half-open cells, on which the convergence theorem is instantiated and the approximations are checked to genuinely change with N .

The discretization structure `DiscreteThermodynamics` carries that predicate as a field, which is what makes its edge weights integrals over a geometrically constrained family rather than over arbitrary sets, with three consequences for the coupling weights. `weight_self`: a vertex has no edge to itself, so the coupling matrix has zero diagonal. `face_of_weight_ne_zero`: a nonzero weight $w(u, v)$ forces $\{u, v\}$ to be an actual face of the complex, so the discretization cannot invent couplings between vertices the triangulation does not join. And `total_weight_eq_setIntegral`: $\frac{1}{2} \sum_u \sum_v w(u, v) = \int_S |T|$, i.e. the total coupling strength is exactly the scalar stress-energy of the region covered — nothing counted twice (disjointness), nothing dropped (covering). That last identity is the precise sense in which the coupling matrix is derived from the continuum rather than posited, and it is exactly what the everywhere-empty triangulation violates.

Examples §6 witnesses the structure on the same grid: each consecutive edge carries weight $1/N$, and the weights sum to 1 for every N . The witness is one-dimensional and its stress-energy tensor is constant, so it establishes satisfiability, not a physically interesting discretization.

One field of the structure is deliberately left free and is marked `[MODELLING]`: `scalar_magnitude`, the scalar read off the stress-energy tensor, constrained only to be non-negative. A $(0, 2)$ tensor has no canonical scalar — the trace, an observer’s energy density and the tensor norm are different functions, and the structure carries neither a metric nor a frame — so imposing one would pin a choice rather than derive it, and the three statements above hold for every non-negative choice, so it would prove nothing new.

That the choice is nonetheless a physical input is stated as a theorem rather than left as a remark: §6 exhibits a second instance over the same triangulation, tensor and regions, differing only in that field, for which the partition identity again holds while every edge weight differs (`gridThermo_magnitude_matters`). The remaining constraint, non-negativity, is the discrete shadow of an energy condition and is what makes the induced couplings a legal Kuramoto coupling.

4.6 The fold-and-cross proof of strict decrease

Uniqueness of the coherent branch is the injectivity of $E(a) = \mathbb{E}_a[\sin^2 \theta]$ on $(0, \infty)$, and what is proved is the stronger statement that E is strictly decreasing on $[0, \infty)$ (`vonMisesSRatio_strictAntiOn`). A folding and crossing argument proves the result without differentiation. Differentiating the exponential family in its natural parameter gives $E'(a) = -\text{Cov}_a(\cos^2 \theta, \cos \theta)$, which costs differentiation under the integral sign and leaves a covariance that is not sign-definite pointwise: symmetrising with an independent copy gives $\frac{1}{2}\mathbb{E}[(X - X')^2(X + X')]$ with $X = \cos \theta$, an integrand changing sign with $X + X'$. The target is kept as the two-point inequality $C_2(a)Z(b) < C_2(b)Z(a)$ for $a < b$, and two moves settle it.

Fold first: the two reflections that proved $I_2 \geq 0$ carry both integrals onto $[0, \pi/2]$, replacing $e^{a \cos \theta}$ by $W(a, \theta) = 2 \cosh(a \cos \theta)$ and confining $\cos \theta$ to $[0, 1]$, after which $\cos 2x - \cos 2y$ and $W(b, x)W(a, y) - W(a, x)W(b, y)$ have the same sign, both being governed by $\cos x$ against $\cos y$. This is the symmetrisation $X \mapsto -X$ that the covariance form needed the tilt to perform, done once at the start; its analytic content is one identity, $2 \cosh X \cosh Y = \cosh(X+Y) + \cosh(X-Y)$ with $\cosh$ increasing in $|\cdot|$.

Then cross: instead of integrating that inequality over a square, let α be the mean of $\cos 2x$ under $W(a, \cdot)$ and $x_0 \in [0, \pi/2]$ a point where $\cos 2x_0 = \alpha$, supplied by the intermediate value theorem; evaluating at $y = x_0$ and integrating in x alone gives the result, since the term carrying $W(b, x_0)$ vanishes by the choice of α . No product measure and no Fubini appear.

Run with $\cos \theta$ in place of $\sin^2 \theta$ the same argument needs no fold — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the variable the family is tilted by — and gives `besselRatio_strictMono`: R is strictly increasing on all of $\mathbb{R}$, the monotonicity of the Bessel ratio.

5 Global Sections and the Unity of Consciousness

Phase synchronization does not imply overlap compatibility. The model represents global consistency by gluing compatible local descriptions; its identification with conscious unity remains a separate stipulation.

Theorem 5 (Sheaf-Theoretic Unity): Given the sheafification of finite spatial measures over a continuous substrate X , local measures that agree on all spatial intersections $U \cap V$ uniquely glue into a global spatial measure. The overlap agreement is an independent physical hypothesis; phase-locking does not imply it. This is not a theorem about arbitrary probability marginals over decoded contents, for which compatible overlaps need not determine a unique global joint distribution Abramsky & Brandenburger (2011).

Lean 4 Implementation (Phase5_GlobalSection.lean): The headline gluing theorem separates two inputs. The class field `ThermodynamicCover.thermodynamic_equilibrium` places the phase field at the potential minimum, while `LocalSectionSynchronization.section_agrees_of_phase_eq` independently requires equal-phase patches to agree on overlaps. Only the second supplies the compatible family consumed by the sheaf theorem; no dynamics in the development derives it.

What the second one says can be stated without the sheaf: on a discrete substrate two patch sections have the same restriction to their overlap exactly when they assign the same mass to every site they share (`restrict_eq_iff_densityOn_eqOn`), so the hypothesis is that patches at a common phase carry a common local density where they meet. It excludes data the rest of the structure admits: a two-patch cover of the three-site substrate at one phase, with profiles differing at the shared site, satisfies every other requirement and fails this one (`overlap_agreement_fails`). It does not require patches to agree away from their overlaps — the three-patch witness carries three pairwise-different profiles.

The equilibrium condition can be obtained from a convergence result for a class of initial data: `Phase5_EquilibriumBridge.lean` supplies `ThermodynamicCover.ofConvergentTrajectory` from `kuramoto_tendsto_global_minimum`, and `Examples §17.1` is a three-patch cover built over the resulting phase field.

The cover class also has an independent instance (`Examples, §4`), so the theorem is not conditional on a structure of unknown realizability. Its three-site substrate has two overlapping patches meeting at one site; local sections are germ families of the phase-indexed measure $\mu_\theta = (1 + \cos \theta)^+ \delta_{\text{mid}}$ in the sheafification of the probability presheaf. The family is provably non-constant in θ (`phaseMeasure_not_const`), and `thermodynamic_equilibrium` follows from `phase_locked_minimizes_potential`.

Two further results specify the phase and section data. The theorem `potential_min_iff_phase_locked` (`Phase4_KuramotoDynamics.lean`) characterizes the minimizers of the reduced potential in both directions, from which `ThermodynamicCover.phase_locked` follows for every instance. The witness in `Examples §13` carries a non-constant phase field and a non-uniform invariant measure, with patch-local densities read from the sections through `densityOn`; agreement at the shared site is an equation between

separately computed numbers. The limit is structural: phase-locking equates phases, while agreement of the associated sections remains the independent class field named above.

The gluing construction assumes local overlap agreement and derives both existence and uniqueness of the global section. The lemma `probability_glue_unique` takes a compatible family to its unique global extension; `ThermodynamicCover.invariantMeasure` denotes that section. The construction uses local data, not a global state supplied as a premise.

5.1 Approximate overlap agreement

Theorem 5 admits no tolerance: the sheaf condition is an equation. Under an ε -tolerance neither existence nor uniqueness of a global section survives, and what replaces them is a selection rule with an explicit constant. The results are stated for a finite substrate whose local states are nonnegative site masses, with the uniform distance $\max_{x \in U} |f(x) - g(x)|$ on the sites a pair of patches shares (`ApproximateGluing.profileDist`, `Phase5_GlobalSection.lean`).

Let $\{U_i\}_{i \in I}$ be a finite cover, let s_i be the local profiles with $\text{dist}(s_i, s_j) \leq \varepsilon$ on $U_i \cap U_j$ for *every* pair (`Compatible`), and let $w_i \geq 0$ vanish off U_i and satisfy $\sum_i w_i(x) = 1$ at every site (`IsPartition`). The selection is

$$(\text{select } w \text{ } s)(x) = \sum_i w_i(x) s_i(x).$$

It depends on the profiles only through their values on their own patches (`select_eq_of_eqOn`), so the bookkeeping values a patch carries elsewhere do not enter. Four estimates follow. The selection is within ε of each patch on that patch (`select_close`); two partitions subordinate to the same cover select states at distance at most ε (`select_dist_le`); any two profiles fitting every patch to tolerance δ lie within 2δ of each other (`approximate_diameter_le`); and at $\varepsilon = 0$ the selection is the unique profile fitting all patches (`select_glue_unique`).

The constant in the second estimate is $C(N) = 1$ at every cover multiplicity N , rather than the $N\varepsilon$ a term-by-term count would give: the weights sum to one rather than to the number of patches active at a site. This is a statement in the uniform mass metric and not in total variation. A feature F that is L -Lipschitz on profiles therefore satisfies $\text{dist}(F(\text{select } w \text{ } s), F(\text{select } v \text{ } s)) \leq L\varepsilon$ (`feature_dist_le`), which is the resolution condition an interpretation at scale δ must meet; the theorem chooses no decoder.

The three-site cortex witnesses both halves (`Examples §21`). Its three two-site patches have multiplicity two, and the family `apxW` is §17.1's data with one patch reporting ε more mass everywhere it can see. Two partitions — each site assigned to the lowest, respectively the highest, indexed patch containing it — select states at distance exactly ε (`apx_select_dist`), so the constant is attained and cannot be improved at this multiplicity. For every positive ε that same family has no exact global section at all (`apx_no_global_section`): two patches see the site `right` and disagree there, so selection stands in for a gluing that does not exist. The corresponding sections fail the class field of Theorem 5 exactly as the two-patch fence does (`apxSection_overlap_fails`), and the tolerance cannot be dropped to zero (`apxW_not_compatible_zero`).

Checking a spanning subfamily of the overlaps is not checking the family. The family `chainW` agrees exactly on the overlaps $(0, 1)$ and $(1, 2)$ and is off by 2ε on $(0, 2)$; its two selections are 2ε apart (`chainW_not_compatible`, `chainW_select_dist_gt`), so the hypothesis quantifies over all pairs. On the same witness the total-mass readout moves by 2ε between the two selections, within the 3ε its Lipschitz constant permits (`totalMass_apx_dist`, `totalMass_apx_dist_le`): the selection dependence of a readout is a real effect and not an artefact of a loose bound. At $\varepsilon = 0$ the three-patch family of §17.1 satisfies the relaxed hypothesis and the selection returns its glued profile (`trioW_compatible`, `trioW_glue_unique`).

The scope is narrow in three ways. The state space is finite spatial mass profiles, where restriction is functorial; nothing here bears on matching marginals of decoded contents Abramsky & Brandenburger (2011). Nothing produces the approximate agreement, selects the cover or selects the weights, so the interpretive cost of the route is that unity becomes a choice at resolution ε . Theorem 5, the `ThermodynamicCover` instances and the composed chain retain their exact agreement hypothesis.

6 Reflexive Topology and the Self

Self-representation is an additional relation between the glued global state and an encoding within it. The model specifies an encoding and readout whose composite fixes that state; contraction gives uniqueness. A dissipation bound imposes neither the encoding nor its accuracy.

Theorem 6 (Conditional Auto-Resonance): On a nonempty complete metric space of global sections, a self-prediction map has a unique fixed point if it is Lipschitz at the independently assumed rate $q = e^{-(K-K_c)\tau/2} < 1$. A state with reconstruction residual at most ε is within $\varepsilon/(1-q)$ of that fixed point. Coherence supplies $K > K_c$; it does not supply the Lipschitz law. To identify this fixed point with the section glued on a given cover, that section must itself be fixed by the map.

6.1 States and their metric

The construction in `Phase6ReflexiveTopology.lean` applies Banach’s theorem to a complete metric space of global sections. The three-site witness in `Examples`, §10, equips these sections with a metric derived from their spatial measures.

On a discrete substrate, every point has a smallest open neighbourhood. Restriction preserves point masses, so a germ at a site determines a mass there. The map `massEquivOn` identifies sections over an open set U with mass profiles on U : injectivity follows because restriction to a point determines the germ, and surjectivity because a finite measure realizes every profile on the finite substrate. The global case is `massEquiv`. This correspondence also constructs the patch-local sections used by the compatibility witnesses.

The distance between two global sections is the largest difference between their site masses. Completeness follows by constructing the section represented by the limit of a Cauchy sequence of profiles. On this finite substrate, the metric is equivalent to total variation within a factor of the number of sites, but is not equal to it. Distinct sections at distance $1/2$ witness its nontriviality (`exists_dist_lt_one`).

The metric matters to the fixed-point argument. Under the discrete $0/1$ metric, every contraction is constant (`contracting_implies_const`). The witness instead uses uniform distance between densities. It makes no claim of equivalence with the Lévy–Prokhorov metric: on a discrete substrate, thickening by $\varepsilon < 1$ leaves a set unchanged, and averaging a measure with a baseline need not contract that metric.

6.2 Encoding, readout and information loss

A `ReflexiveBoundary` carries an encoding into a local region, called the avatar, and a readout back to global sections. Its predictive map is their composite. This ties reconstruction to the information passing through that region. A map that reads the whole field cannot serve as a witness unless it factors through the stated local encoding.

The separate predicate `IsRestrictionResonance rb` requires the encoding to be presheaf restriction to the avatar region. Under this condition,

$$F(s) = \text{readout}(s|_U), \quad s_* = \text{readout}(s_*|_U)$$

at a fixed point. The second equality is `self_eq_readout_restrict`. The more general `self_mem_range_readout` places a fixed point in the range of the readout even without restriction resonance. These statements express what the local encoding reconstructs; restriction resonance is an assumption on the boundary, not a consequence of contraction.

Local readings can also determine a global state collectively. If a family of resonant avatar regions covers the substrate, equal readings at every avatar imply equal global states (`eq_of_avatar_eq`). The proof uses the same gluing uniqueness as Theorem 5. A single avatar can discard information away from its region, so collective reconstruction from a cover must be distinguished from reconstruction through one region at a fixed point.

6.3 The contraction hypothesis

For $\tau > 0$, the assumed factor is

$$q = \exp\left(-\frac{K - K_c}{2}\tau\right), \quad K_c = 2D.$$

The lemmas `resonanceRate_lt_one` and `one_le_resonanceRate` split the parameter plane at the threshold. Above it, a map Lipschitz at this factor has a unique fixed point (`self_of_supercritical`). The bridge `self_of_coherent_order_parameter` obtains the required inequality from the existence of a coherent order parameter. At or below threshold, the specified factor fails the strict bound required by `ContractingWith (not_contractingWith_resonanceRate)`; this argument is unavailable there.

The rate is motivated by the mean-field linearization near the incoherent state, $\dot{r} = \frac{1}{2}(K - K_c)r + O(r^3)$. Linearizing the cubic normal form about its coherent branch gives relaxation rate $K - K_c$, twice as fast. The assumed factor q is therefore larger than the coherent-branch relaxation factor. Both cross one at $K = K_c$. This comparison motivates a choice of bound; it does not derive a Lipschitz law for a neural encoding or readout from phase dynamics.

Banach's theorem establishes uniqueness for the specified map. E89 additionally requires the section glued on the specified cover to be fixed by that map, making it the unique fixed point. The theorem `chain` uses this equality and the contraction law but does not require restriction resonance. The interpretation through $s|_U$ needs that additional predicate.

6.4 Approximate reconstruction

Let $F(s_*) = s_*$ and $d(s, F(s)) \leq \varepsilon$. The triangle inequality and the Lipschitz law give

$$d(s, s_*) \leq d(s, F(s)) + d(F(s), F(s_*)) \leq \varepsilon + q d(s, s_*),$$

hence $d(s, s_*) \leq \varepsilon/(1 - q)$. The theorem `ReflexiveBoundary.approximate_self_bound` applies Mathlib's `ContractingWith.dist_le_of_fixedPoint`. An estimate about a supplied fixed point requires neither completeness nor a separate existence proof; those hypotheses enter the existing Banach construction. The approximate states can form a set while the exact fixed point remains unique.

If $d(t, s) \leq \delta$, the same fixed map satisfies

$$\begin{aligned} d(t, F(t)) &\leq \varepsilon + (1 + q)\delta, \\ d(t, s_*) &\leq \frac{\varepsilon + (1 + q)\delta}{1 - q}. \end{aligned}$$

These are `prediction_residual_of_dist_le` and `approximate_self_bound_of_dist_le`. The errors add before division by the contraction gap, whose inverse amplifies the bound near threshold. The displacement bound is supplied; this result does not derive approximate gluing or weaken Theorem 5's overlap hypothesis.

For $K > 2D$, $\tau > 0$ and $\varepsilon \geq 0$, `approximate_self_bound_of_supercritical` instantiates the estimate at the prescribed rate, and `resonance_error_radius_bounds` proves

$$\frac{2\varepsilon}{(K - 2D)\tau} \leq \frac{\varepsilon}{1 - e^{-(K-2D)\tau/2}} \leq \frac{2\varepsilon}{(K - 2D)\tau} + \varepsilon.$$

The proof uses $1 - e^{-x} \leq x$ and $x/(1 + x) \leq 1 - e^{-x}$ for $x = (K - 2D)\tau/2 > 0$, both consequences of $1 + x \leq e^x$. Thus the radius has leading behaviour $2\varepsilon/((K - 2D)\tau)$ and diverges as $K \downarrow 2D$ at fixed positive ε and τ . This concerns the tolerance bound, not an actual state's error. Zero residual still gives zero distance at every supercritical coupling. The map, metric and Lipschitz law remain specified inputs; no result here models a changing readout or constructs a biological encoding.

6.5 A nonconstant witness and its controls

On the three-site cortex, the avatar is one site. The encoding reads its mass (`avatarRead`), and `avatarReadout` reconstructs a profile by averaging the reading with a baseline. The resulting map is nonconstant, has a unique fixed point, and contracts by one half of the difference visible at the avatar (`cortexPredict_not_const`, `cortexPredict_fixed_unique`, `cortexPredict_dist`). Differences elsewhere are invisible to this map. The whole-field averaging map `relax` therefore cannot replace it as a witness for a one-site encoding.

The witness takes $K = 3$, $D = 1$ and $\tau = 2 \ln 2$, so $q = 1/2$ (`cortexResonanceRate`). These are parameters chosen for the example, not calibrated cortical values. The boundary `cortexReflexive` performs restriction and satisfies resonance. Its avatar distinguishes the silent field from the baseline, as shown by `cortexReflexive_restrict_ne`. One avatar per site gives a covering family and reconstruction from the three readings (`cortex_eq_of_avatar_eq`). The same map at $K = 1$ exercises the failure of contraction at the prescribed subcritical factor (`cortexSubcritical_not_contracting`).

Replacing the encoding by a constant makes the composite constant. It still has a unique fixed point, namely the readout of that constant (`constResonance_existsUnique_self`). In the cortical control, this fixed point differs from the original baseline (`cortexBlind_self_ne`), and distinct states have identical blinded readings (`cortexBlind_not_determined`). A constant encoding fails restriction resonance

whenever two states differ on the avatar region (`constResonance_not_isRestrictionResonance`). The control thus distinguishes fixed-point uniqueness from sensitivity to the encoded state.

The approximate bound is sharp on the same one-site-avatar witness. The silent profile has $d(s, F(s)) = 1$ and $d(s, s_*) = 2$, attaining $\varepsilon/(1 - q)$ at $\varepsilon = 1$, $q = 1/2$ (`cortexSilent_prediction_residual`, `cortex_approximate_self_sharp`). At $q = 1$, `cortexIdentity` instead takes the whole substrate as its avatar and uses identity encoding and readout. It is nonexpansive, and the silent and baseline profiles are distinct fixed points (`cortex_nonexpansive_not_unique`). This control establishes the failure of nonexpansive uniqueness on the same metric; it is not a model of compression into a proper local region.

6.6 What a biological interpretation must add

The readout, its contraction law and the restriction-resonance condition are choices or hypotheses of a model. No field dynamics here produces them. The squared-drift objective is absent from the fixed-point result, so its minimizer is not identified with the self-model's fixed point.

The article's cup example illustrates the further content assignment: the represented state would include the scene and the perceiver's relation to it, and the encoding would need a demonstrated role in representing that relation. The three-site witness establishes mathematical realizability, not such a role. A nonconstant reconstruction and a unique fixed point alone do not establish a perspective or experience. Interventions on a candidate neural encoding would have to test the functional interpretation separately from the equality defining its fixed point.

7 Coupling support and continuum observables

Two mathematical comparisons distinguish a restriction on wiring support from a restriction on the measure used to evaluate a kernel.

Theorem 7 (Resource-matched coupling reallocation): If a fixed wiring support excludes a pair whose phase correlation exceeds the maximum over its wired pairs, a coupling using that pair can achieve greater total correlation at the same resource budget. This is a conditional support comparison, not a general inequality between biological and silicon hardware.

Coupling reallocation (Phase7): The lemma `exists_better_coupling_allocation`, wrapped as the theorem `continuous_beats_rigid_topology`, proves the reallocation step: given a coupling matrix satisfying `is_strictly_suboptimal` (positive weight on a less phase-correlated pair while a more correlated pair exists), there is another valid coupling with identical total resources and strictly greater total phase correlation.

Lean 4 Implementation (Phase7.Rigidity.lean): Two results specify the scope of the comparison. (i) *Rigidity as a support constraint.* `RealizableIn S R A` says A is a valid coupling meeting resource budget R and vanishing on every pair the architecture has no wire for — weights tunable, wiring fixed. `totalCorrelation_le_of_realizable` proves no realizable coupling exceeds $R \cdot M$ when M bounds phase correlation over the wired pairs, and `exists_realizable_pair` attains it. `rigid_is_strictly_suboptimal` then *derives* `is_strictly_suboptimal` for every realizable coupling, whenever some pair (k, l) the wiring misses has correlation exceeding M : the hypothesis Theorem 7 would otherwise have to assume is a consequence of the wiring pattern. `rigid_gap` gives the quantitative form, a gain of at least $R(\cos(\theta_l - \theta_k) - M) > 0$ at equal budget. `Examples §8` witnesses this on three sites wired only to the maximally anti-correlated pair, where the gap is exactly 2, and runs `exists_better_coupling_allocation` on it with no assumed hypothesis. (ii) *Continuity as a measure-theoretic distinction.* `fieldCorrelation` integrates a kernel against $\mu \otimes \mu$ as in Derivation 4. A finitely-sited architecture is a `Finset X`; `fieldCorrelation_sited_eq_zero` proves that on an atomless substrate such a kernel contributes exactly zero, for every phase field and every choice of weights, while `fieldCorrelation_patchKernel` gives $c\mu(U)^2$ for a patch. `sited_architecture_below_field_optimum` combines them; `Examples §8` instantiates it on $\mathbb{R}$ with a thousand units of weight at the origin against the unit interval.

Scope caveat. Result (i) is resource-matched and not question-begging, but it turns on fixed wiring support, not on continuity: an architecture wired to the best pair is not beaten. Result (ii) is a genuine type-level separation but is *not* resource-matched, since a measure-zero substrate carries zero resource in the $\mu \otimes \mu$ sense as well; what it establishes is that the continuum coupling energy cannot register a finitely-sited architecture at all, which is a claim about the model's observable rather than about computational power. Neither result formalizes manifold structure or lattice geometry. Neither result establishes that digital hardware cannot approximate field dynamics or support consciousness.

8 Claims and their Lean identifiers

The table below is the claim-by-claim map, kept here rather than in the main text. It is here because it is a reference rather than an argument: each row names the Lean identifier that carries a claim and states the scope limitation of that identifier, and no step of the main text depends on the reader working through it. Figure 1 in the main text separates the three requirements and their inputs; Table 1 records the eight connecting hypotheses of the formal composition. The present table records node-level results and their implementation scope. The cover’s compatibility is a supplied field. The boundary’s restriction-resonance condition is a separate predicate, needed for the interpretation of self-encoding as restriction but not consumed by **chain**.

Table S1: Mapping of physical claims to Lean 4 formalization status. The development declares **no axioms**: `#print axioms` on any result above reports only `propext`, `Classical.choice` and `Quot.sound`. A physical postulate is an obligation on each instance rather than a global assertion, which is the shape a standalone axiom cannot have (Section 9); two of the postulates a reader might expect to find are not postulates at all, but consequences of the local data (Derivation 5). “Theorem” still does not mean “physically established”: the postulates live in hypotheses and instance obligations, and every such class is inhabited by a witness in `Examples`, so no row is vacuous.

Conceptual Claim	Lean 4 Implementation	Status
Continuing finite agent	<code>FiniteProtocol</code> sequences different operations on one evolving law. <code>ContinuingProcess</code> bounds total heat and joint entropy reduction by a declared work store plus the system’s mean energy drop. <code>ContinuingAgent</code> executes actuation, observation-driven learning and reset; <code>Examples/ContinuingAgent.lean</code> proves persistent above-chance reward, positive recurring cost, a funded first cycle and exclusion of 22 cycles at its declared allowance. <code>Blinding</code> removes the reward advantage; omitting reset reduces later reward. <code>Chain.activeBound.of.continuing</code> derives stage budgets when every stage has nonnegative mean heat. The store constrains expectations at every prefix, not individual paths; the prior, store and composite-channel reservoir model are supplied. Sensor-memory implementation, battery dynamics, fabrication, register-ledger identification and cortical interpretation remain separate	Theorem (conditional)
Finite goal-directed control	<code>FiniteControlProblem (Phase3.AgencyThermodynamics)</code> evaluates terminal reward and both substep costs on the same <code>FiniteFeedbackCycle</code> , with an explicit deterministic policy and shared world and memory channels. <code>exists.optimal</code> proves finite feasible maximization. <code>Examples/AgencyControl.lean</code> computes every deterministic bit policy’s performance, heat and work and proves a unique feasible optimum that strictly improves a declared lamp task. A feasible baseline misses the target; a better-performing policy exceeds the budget; equal final laws can have unequal heat. Objective, reservoir model, initial resource and budget are supplied. This comparison covers fixed policies; adaptation of a policy register is treated separately below	Theorem (conditional)
Finite learning dynamics	<code>FiniteFeedbackStep.iterate</code> carries successive laws and telescopes entropy and first-law balances. <code>Examples/PolicyLearning.lean</code> proves strictly improving expected lamp reward, geometric convergence to a stationary mixture, positive update heat and bounded cumulative heat from the register’s energy loss. Its energy encodes known task values. The limit is suboptimal; individual updates and a different initial preparation can lower reward. Task costs are prospective lamp task expectations. Preparing the register, executing task episodes, implementing the policy readout, cortical identification and the active chain’s resource and spatial bridges require separate accounting	Theorem (conditional)

Conceptual Claim	Lean 4 Implementation	Status
Learning from observed outcomes	<code>FiniteObservationalLearner</code> (<code>Phase3.ObservationalLearning</code>) holds an environmental parameter fixed and updates a policy register from the observed outcome of the action its readout executed. Neither the update nor the readout takes the parameter as an argument and the reward enters no channel, so nothing supplies the task's values. Act-observe-update composes into one <code>FiniteFeedbackStep</code> , carrying reward, heat and entropy on the same law. <code>Examples/ObservationalLearning.lean</code> gives every mass of that law at every horizon for a two-action task with an unknown rewarding action, hence strictly improving expected reward $3/4 - \frac{1}{4}(1/2)^n$ from a uniform prior, strict improvement over a frozen register and over an uninformative world under a common allowance, cumulative heat $(\log 3/4)[1 - (1/2)^N]$ and its equality with external work. <code>zero_work_needs_parameter_independent_heat</code> proves that a free relaxation would carry no information about the parameter, so the work is irreducible. The limit is below certainty, each action is executed half the time at every horizon, and a reward-decreasing transition keeps positive probability. Preparing the prior, supplying the work, task episodes beyond the declared updates, fabricating the readout, optimal-policy convergence and cortical identification are separate	Theorem (conditional)
Actuated scalar coupling	<code>ActuatedCoupling</code> fixes density $g[H(p) - H(p')]b(x)$ for a named feedback step. Regular mesh refinement proves convergence; <code>Chain.actuated_limit_le_budget</code> bounds its limit by $g(B/\theta) \int b$ when the total profile is nonnegative. A noisy actuator has positive drive and changing spatial energies; stationary execution, changed gain and changed profile alter the result. The constitutive law is supplied and produces no kernel or microscopic actuator work account	Theorem + instance postulate
Microscopic actuation and kernel refinement	<code>LocalActuator</code> (<code>Phase3.LocalActuator</code>) sets a product-space kernel $\sum_i c_i(z)\phi_i(x)\phi_i(y)$ and a stored energy $\sum_i u_i c_i(z)$ from the same mode occupancies. <code>kernel.update</code> and <code>kernel.update.local</code> give a transition's change and its spatial locality; <code>expected_kernel.update</code> averages those pathwise changes over the executed step, and <code>installation.first.law</code> charges the same paths with the stored-energy change plus the channel's heat. <code>KernelMesh</code> (<code>Phase2.KernelMesh</code>) reconstructs each sampled weight throughout its cell pair, <code>energy.eq.integral</code> identifies the cell-mass weighted energy with that reconstruction's product-measure integral, and <code>energy.tendsto</code> converges it under shrinking sample error on a compact substrate. <code>Chain.e45Active_of_localActuator</code> and <code>Chain.e45_of_localActuator</code> discharge the two spatial edges for an arrangement holding, respectively, a named step's final law and a declared predictive system's joint law; neither proof consults its premise. A thermal switch installs one mode for positive external work, changes a pairwise value, changes its coupling energy between the first two partitions of an atomless interval, and rejects a zero-work and a wrong-limit claim. Profiles, prices, readout, reservoir and substrate are declared hardware; fabrication, continuing power and cortical identification are not established	Theorem + instance postulate
One register's resource ledger	<code>RegisterLedger</code> (<code>Phase3.AgencyThermodynamics</code>) carries one register's update, its operations as elementary steps controlled by that register, local detailed balance for each at that register's temperature, and the identity that its dissipated heat is the total those operations deliver to its reservoir. <code>opHeat.nonneg.of.compressive</code> is the path-model second law at that temperature and <code>opHeat.le.dissipation</code> derives the allocation; <code>ledger.bathEntropy</code> exhibits the total as the register's own bath entropy change. <code>Examples/RegisterBudget.lean</code> witnesses it on the one-bit eraser with two operations delivering $(2/5)\log 2$ and $(3/5)\log 2$, and fences both inputs: a compressive operation above the whole dissipation, and a ledger whose other operation draws heat out of the reservoir. The identity and the compressiveness are physical inputs; sequencing the operations into one evolving law, a continuing energy source and cortical identification are separate models	Theorem (conditional)

Conceptual Claim	Lean 4 Implementation	Status
A register's microscopic bath	Examples/RegisterBath.lean realizes each of the four maps of a register bit by a permutation of register and bath acting on a bath prepared pure, and computes heat as the bath's mean energy gain on the executed paths: $\log 2$ for the erasures, 0 for the injective maps, inhabiting Landauer's heat equation with the reachable bath supports. sum.energyTransfer and reported.heat.eq (Phase3.ContinuingAgent) give this ledger for any finite protocol with no positivity. Three results fence a map-indexed budget: the same gate run twice restores the prepared law, with shares $\log 2$ and $-\log 2$ about a total of zero; a drive that holds the register fixed excites its bath at $2 \log 2$, so the map does not fix the cost, and the work is $\log 2$, $2 \log 2$ and zero for the three runs; and a deterministic gate is not strictly positive, while an involution's log-ratio heat vanishes on every realized path though its bath gains $\log 2$. Positivity, local detailed balance and the ledger identity are therefore inputs rather than consequences of reversibility. The instance is local to its file; preparation, work supply, fabrication and identification of this bath with the ledger's reservoir are separate	Theorem (negative)
Agency and feedback thermodynamics	Phase3.Agency composes policy, world, observation and memory-update channels. It proves the signed conditional-information identity, conditional nonnegativity under finite information, and passive recovery. Phase3.FiniteInformation connects finite laws to the same KL definition and proves finite-state finiteness. Phase3.AgencyThermodynamics derives the joint entropy balance from normalized positive path laws; local detailed balance supplies heat. Its FiniteFeedbackCycle constructs the swapped intermediate law, identifies the composed channel output and proves total entropy, heat, first-law and budget relations at a common thermal scale and energy observable. Reciprocal and biased two-bit witnesses check nontrivial updates and reject a mismatched intermediate law. Chain.ActiveBound carries the named elementary process's entropy, heat and budget into chain.active ; the cycle bound supplies neither the register allocation nor coupling convergence. Extra registers, protocols, learning and cortical readouts require further models	Theorem (conditional)
Capacity of a finite phase space	The first premise, discharged by theorem rather than by footnote (Phase1.PhaseSpaceCapacity). $H(p) \leq \log X $ for any distribution on a finite state space, with equality exactly at the uniform law (shannon.entropy.eq.log.card.iff); a stream with $ P ^k > X $ histories cannot be recorded injectively (absorb.not.injective) and its entropy exceeds anything the system can hold; and on a <i>finite</i> space an unreachable state is an erasure, hence heat (finite.phase.space.dissipates). Finiteness is assumed, not derived, and is fenced — Nat.succ refutes the last step on an infinite space (Examples §19). The symmetry group and the metric are still consumed by no theorem	Theorem
Pointwise Vacuum Minimization	pointwise.vacuum.of.global.min (Phase1), potential energy defined as an integral; holds a.e., and a double-well witness proves the a.e. cannot be strengthened	Theorem
Landauer's Principle	Derived from statistical mechanics (Phase3); the axiom-free reversibility core becomes a heat bound via the field StatisticalMechanics.heat.eq	Theorem + instance postulate
Symmetry Breaking implies Boundaries	Derived from pointwise potential minimization (Phase1); witnessed on a double well with a $\mathbb{Z}_2$ symmetry whose minimisation hypothesis is proved, not assumed	Theorem
Inevitability of Boundaries (π_0)	The domain-wall case of the Kibble mechanism (Phase1 §3): on a connected substrate, a continuous field taking values in two different connected components of the vacuum manifold must leave the vacuum somewhere (exists.notMem.of.connectedComponentIn). Witnessed on the double well, whose vacuum manifold is computed to be exactly $\{-1, 1\}$ and proved disconnected (Examples §11.1); the field class is non-empty and the forced wall is located. Higher homotopy (π_1 vortices, π_2 monopoles) available and not taken	Theorem

Conceptual Claim	Lean 4 Implementation	Status
Nonpredictive information bounds dissipation	Still, Sivak, Bell and Crooks' bound is the field <code>PredictiveDissipation.still.bound</code> (Phase3_ <code>PredictiveThermodynamics</code>). Everything downstream is derived: mutual information is defined from Mathlib's <code>klDiv</code> ; the data processing inequality for the chain $X_t \rightarrow S_t \rightarrow S_{t+1}$ proves the nonpredictive information non-negative (<code>predictiveInfo.le.mutualInfo</code>), from which $W_{\text{diss}} \geq 0$ follows rather than being assumed; a dissipation budget bounds wasted memory, and zero dissipation forces every retained bit to be predictive. Witnessed in Examples §18 by two systems over one two-bit law, one permitted to dissipate nothing and one forced to dissipate its whole memory, with a counterexample showing the no-back-action hypothesis is load-bearing. Descent of σ alone does not imply $D_{KL} \rightarrow 0$. On a finite register with a bipartite environment the postulate is discharged rather than assumed: <code>PredictiveDissipation.ofLandauer</code> (Phase3_ <code>LandauerBridge</code>) builds a predictive structure whose thermal scale and dissipated work are the register's own and derives <code>still.bound</code> from <code>landauer.bound</code> , given the identification of wasted memory with erased entropy. That identification is the physical content and is not free — a reversible update erases nothing, so it forces the waste to zero (<code>nonpredictive.eq.zero.of.injective</code>) — and on the one-bit eraser the bound is attained (Examples §18.5)	Theorem + instance postulate
Structural Resonance KL Bound	Gibbs' inequality proved; the bound is the field <code>StructuralResonance.kl.bound</code> (Phase3_ <code>KLBound</code>). The postulate is consistent and yields $\sigma \geq 0$. No limit may be read off it	Theorem + instance postulate
Fixed-phase kernel descent	The bounded drift operator and its Hilbert-space adjoint give the squared-drift gradient. The theorem <code>structural_resonance.decreases.sigmaContinuum</code> (Phase8) proves descent for the specified flow with phases fixed. An atomless witness is supplied in Examples, §20. Neither joint dynamics nor thermodynamic entropy production is established	Theorem
Phase-locking minimizes squared drift	For a <i>reciprocal</i> coupling kernel, the phase-locked field has the least squared-drift objective of <i>any</i> field (<code>phase.locked.minimizes.entropy.of.symm</code> , Phase8 §2a). The mean-matching condition is derived: the coupling term is antisymmetric under exchanging its two sites, so total drift is conserved across phase fields (<code>total.drift.eq.of.symm</code>). Reciprocity is necessary — a one-way coupling on two sites strictly beats its own phase-locked state (Examples §16)	Theorem
Electrodynamic Discretization	Lebesgue-Bochner edge weights (Phase2) sitting on faces of the complex and partitioning the region's stress-energy; mesh refinement converges, witnessed on $[0, 1)$. The scalar read off the stress-energy tensor is a marked modelling choice: the partition holds for any non-negative choice, and two instances differing only in it have different couplings (Examples §6)	Theorem + instance postulate
Coarse-graining of a symmetric coupling	<code>mesh.refinement.convergence</code> (Phase2) proves convergence of the scalar coupling energy along regular refining triangulations. On the formalized $[0, 1)$ grid the energy is the composite trapezoidal rule, so a C^2 integrand has error at most $\zeta/(12N^2)$. The result produces neither a kernel nor dynamics: <code>edge.region</code> supplies no product-structured data on $M \times M$. A kernel limit needs a different discrete architecture, which <code>KernelMesh</code> supplies through cell pairs rather than edges; the dynamical mean-field problem stays separate from both	Theorem (scalar energy only)
The field is the endogenous EM field	Conditions formalized; the identification itself not derived. A continuum coupling kernel is posited after the scalar convergence result; identifying that kernel with cortical ephaptic coupling is the framework's central empirical commitment (the physical-calibration subsection of the main article). <code>IsEMFieldCoupling</code> (Phase9_ <code>EMIdentification.lean</code>) is the predicate <code>E56</code> carries: a jointly continuous kernel, a normalised substrate, a coupling constant K equal to that kernel's double average, no single site contributing more than half of it, and a noise D that is the field's own. No object in this development denotes cortex, so whether cortex satisfies them is not a Lean question. Supported as <i>modulatory and integrative</i> , not as primary: measured endogenous fields of 1–5 mV/mm shift spike timing by 1–3 ms. Calibration and observational tests are stated in the main article	Modelling assumption + empirical commitment
Frustration as memory capacity	Outside the regime the chain covers. <code>ThermodynamicCover.A.pos</code> excludes frustrated couplings at the gluing node, and circle-valued offsets from patch phases are coboundaries. Cortical phase singularities supply integer transitions between local real lifts instead: <code>loopWinding.eq.zero.of.hasGlobalLift</code> proves non-zero winding obstructs a global lift, with a winding-one witness. No theorem connects winding sectors to content	Theorem (topological obstruction) + empirical candidate

Conceptual Claim	Lean 4 Implementation	Status
Phase synchronization to Unity	Sheaf-theoretic gluing (Phase5) over a <code>ThermodynamicCover</code>, whose fields are all local: the global section is constructed by <code>probability.glue.unique</code>, and the statement that every local section is its restriction, <code>sync.to.section.eq</code>, is a theorem about it. Witnessed by two patches built from different mass profiles, each section constructed over its own patch, whose glued state is neither of them (Examples §14). The cover's equilibrium hypothesis is not an assumption on every instance: <code>ThermodynamicCover.ofConvergentTrajectory</code> (Phase5_ <code>EquilibriumBridge</code>) discharges it from <code>kuramoto.tendsto.global.minimum</code>, and Examples §17.1 is a three-patch cover whose phase field is the limit of a trajectory started <i>unsynchronized</i>. The overlap-agreement hypothesis is untouched by any dynamics and is read pointwise: two patch sections have the same restriction to their overlap exactly when they assign the same mass to every shared site (<code>restrict.eq_iff.densityOn.eqOn</code>), and a two-patch cover at one phase with profiles differing at the shared site fails it while meeting every other requirement (<code>overlap.agreement.fails</code>, Examples §20). Under an ε-tolerance on overlaps the sheaf condition is replaced by a selection rule: normalized subordinate weights select a state within ε of every patch, two such selections differ by at most ε at any cover multiplicity (<code>select.dist.le</code>), zero tolerance recovers the unique gluing, and for every positive ε a three-patch family at that tolerance has no global section at all (<code>apx.no.global.section</code>, Examples §21)	Theorem + instance postulate
Reflexive Topology (the Self)	Banach fixed point and residual bound $d(s, s_*) \leq \varepsilon/(1 - q)$ for a self-prediction map (Phase6); metric, completeness and the Lipschitz law at $q = e^{-(K - K_c)\tau/2}$ are hypotheses, discharged on the three-site witness. Displacement errors add before division by $1 - q$; the witness attains the bound, and nonexpansiveness alone permits distinct fixed points. Coherence yields $K > K_c$; it does not prove the Lipschitz law. E89 supplies that law and identifies the glued state with the fixed point	Conditional theorem + modelling postulate (witnessed)
Kuramoto Lyapunov descent	Descent chained to the minimum characterization by the rotating frame (Phase4_ <code>RotatingFrame</code>); identical frequencies	Theorem
Kuramoto dynamics run	The initial value problem is well posed on all of $\mathbb{R}$: the field is globally Lipschitz and globally bounded (<code>kuramotoField.lipschitz</code>, <code>kuramotoField.norm.le</code>), and through every state at every instant passes exactly one trajectory, defined for all time (<code>kuramoto.cauchy.problem</code>). Along <i>every</i> trajectory of a zero-frequency system the dynamic potential converges (<code>dynamic.potential.tendsto</code>) and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite, equal to the potential's total drop (<code>dissipation.integral.tendsto</code>). The motion stops: Barbalat's lemma is proved here, Mathlib lacking it, and applied, so $\sum_i \dot{\theta}_i^2 \rightarrow 0$ along every trajectory (<code>velocity.sq.tendsto.zero</code>). Where it stops is proved under an initial-data condition: if the phases start within a quarter turn and the potential excess starts below $a/2$, where $a > 0$ bounds the coupling below, the trajectory converges to a phase-locked configuration that is a <i>global</i> minimiser, with $r^2 = 1$ (<code>kuramoto.tendsto.global.minimum</code>), by a Polyak–Łojasiewicz estimate rather than a LaSalle principle. Unconditionally this is false — splay and twisted states are equilibria — so the condition cannot be removed. Witnessed on three sites, where no closed form exists (Examples §17)	Theorem (conditional)
Continuous vs. Rigid Topology	Fixed wiring support at matched resource (Phase7_ <code>Rigidity</code>), deriving the suboptimality Phase 7 assumed; typed continuum/discrete separation	Theorem
Critical Coupling Threshold ($K_c = 2D$)	Self-consistency equation (Phase8_ <code>SelfConsistency</code>): at or below K_c, threshold included, only $r = 0$ solves it; above K_c <i>exactly one</i> $r > 0$ does (<code>supercritical.fixed.point.existsUnique</code>), so the non-negative solutions are $\{0\}$ below and $\{0, r\}$ above; that r tends to 0 as $K \downarrow K_c$ and increases strictly with K (<code>coherent.branch.strictMono</code>); that it does so at the mean-field rate $r^2/(K - K_c) \rightarrow 1/D$, i.e. $\beta = 1/2$ (Phase8_ <code>CriticalExponent</code>: <code>coherent.solution.critical.exponent</code>, witnessed by <code>coherentBranch.critical.exponent</code>); a solution is a density reproducing its own order parameter. Conditional on the von Mises stationary density, which is an input, and on no link to a trajectory	Theorem (conditional)

Conceptual Claim	Lean 4 Implementation	Status
Conditional composition	<code>Chain.lean</code>: <code>chain</code> and <code>chain.active</code> each take eight named hypotheses, a <code>ThermodynamicCover</code> and a <code>ReflexiveBoundary</code>, and conclude <code>UnifiedSelf</code> through the shared downstream proof. <code>E89</code> supplies the Lipschitz law and identifies the particular glued section with a fixed point; Banach supplies uniqueness. <code>E34</code> requires a passive predictive structure with the register's temperature, heat and nonzero waste. <code>E34Active</code> supplies physical premises and a heat allocation for the named finite feedback process; its entropy bound is derived, and <code>e34Active.ofLedger</code> derives the allocation itself inside one register's resource ledger. <code>E45</code> is discharged for named hardware holding a declared predictive system's own joint law. <code>E45Active</code> is discharged for a scalar actuation law with supplied gain, profile and regular meshes, whose limit is bounded using the named process's heat budget, and for a microscopic arrangement whose product-space kernel is built from occupancies that step changes. <code>E78</code> requires the supplied cover to be reached by relaxation at coupling at least K. <code>chain.hypotheses.jointly.satisfiable</code> and <code>chain.active.hypotheses.jointly.satisfiable</code> witness the two branches with distinct toy components. The active witnesses reject a zero heat budget, an incompatible energy limit, a compressive operation above the register's whole dissipation and a ledger whose other operation draws heat out of the reservoir. Joint satisfiability establishes no common microscopic mechanism; the actuated witness ties the scalar density to the process through a supplied constitutive law and the microscopic witness ties a kernel to it through declared hardware, while mesh regularity supplies convergence in both. <code>E12</code> is equivalent to its conclusion, so the first two links are independent roots. The supplied structures' physical fields and the representational assumptions remain additional inputs	Conditional theorem with eight named hypotheses
Coarse-graining produces no kernel	Negative result (<code>Chain.lean</code> §9). <code>ContinuousNeuralField</code> carries no conditions, so a constructor from a mesh limit would be free of content (<code>continuousNeuralField.free</code>). <code>mesh.refinement.convergence</code> converges a <i>scalar</i>, and <code>TriangulatedManifold.edge.region</code> is a subset of M rather than of $M \times M$, so the discrete side has no product structure to pass to a limit. The vertex-sited candidate does carry the weights (<code>vertexKernel.apply.embedding</code>) and still contributes exactly zero continuum coupling energy on an atomless substrate (<code>vertexKernel.fieldCorrelation.eq.zero</code>) — Derivation 8's hardware no-go, cutting against the framework's own coarse-graining step. The obstruction is to this architecture and not to kernels: <code>KernelMesh</code> carries cell pairs of positive mass rather than vertices, and its limit is a kernel integral	Theorem (negative)

9 Formal soundness: why a global axiom is the wrong shape for a physical postulate

Formalization is only as trustworthy as the axioms it rests on, and an axiom that is inconsistent silently makes every theorem downstream of it vacuous without changing whether the remaining statements type-check. The failure mode below is easy to reproduce: a declaration can constrain a symbol it does not itself bind, making the resulting class inconsistent.

The failure has one shape, and naming it is more useful than cataloguing its instances: *an axiom that constrains a symbol it does not itself bind*. Such a declaration is not a postulate about one system. It is a constraint on every model of the class the symbol belongs to, including models built to violate it, and a class that has such a model is thereby inconsistent. Three concrete cases, each of which proves `False` on its own:

1. `spontaneous_symmetry_breaking_pointwise_min` would take the potential-energy functional as an *implicit* argument occurring only in its hypothesis `PotentialEnergy phi = V v0`. Instantiating that functional with the constant map $\lambda _ . V(v_0)$ discharges the hypothesis by `rfl` and yields the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$ this proves $1 = 0$.
2. `landauer_heat_eq` would equate the free class field `heat.dissipation` with a bath-entropy difference *for every instance* of the class. An instance with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.

3. `kl_bound_axiom` would assert $\sigma \geq D_{KL}(P\|Q)/\Delta t$ for *all* P and Q , while σ is a single real fixed by a class field. Since D_{KL} is unbounded above, a point mass against a uniform distribution refutes it.

Two further declarations of the same shape are worth recording because their refutation is different in kind. `phase_invariant_periodic` and `sync_to_section_eq`, as global assertions pinning free fields of `LocalSectionSynchronization`, are refutable as soon as the probability presheaf admits two distinct global sections; the refutation needs no arithmetic at all, only that the sheaf is not trivial. Neither is a postulate in the current development: the periodicity condition is an obligation of the constructor that builds a cover from an invariant measure, and `sync_to_section_eq` is a theorem about the global section the sheaf condition produces.

The rule that prevents it is uniform: *a physical postulate that mentions a class field must be a field of that class, never a standalone axiom quantified over all instances*. A postulate is then an obligation each model discharges, not a global claim about every model. Under that rule the development declares no axioms at all.

9.1 What the witnesses establish, and what they do not

Explicit witnesses establish that the class assumptions can be satisfied. The `Examples` witnesses therefore cover every class that carries a physical postulate: a one-bit erasure system with a genuine bath that discharges `heat_eq` and reproduces $\Delta Q = k_B T \ln 2$; a perfectly resonant system discharging `kl_bound`; and a scalar field on a one-point spacetime discharging the action principles. The gluing structures are the most demanding case and are witnessed by a `ThermodynamicCover` over a three-site substrate carrying the discrete topology and its Borel σ -algebra, covered by two overlapping open patches whose intersection is a single shared site.

What the witnesses establish is that the classes are inhabitable, so the theorems about them are not vacuous. What they do not establish is that a cortex is such an instance. In the cover above the phase field is already constant — every patch locked at $\theta = 0$ — which is precisely what makes the equilibrium obligation dischargeable; getting a physical system into that configuration is the content of the informal dynamical argument.

9.2 Joint phase and coupling plasticity control

This control tests alignment between a coupling kernel and a specified environmental template. It does not measure overlap compatibility. Its role in the article is to show why coherent adaptation under an objective does not itself supply the representational structure that gluing requires.

A finite driven control co-evolves phases and symmetric nonnegative couplings at fixed total coupling. It uses $N = 100$, $D = 0.1$, $dt = 0.01$, mean row coupling 4, and three nearly equal frequency clusters at 0.5, 1 and 1.5 per unit time. Couplings update every 50 fast steps with learning rate 0.2 for 400 time units, across three seeds. The symmetric-edge derivative of $\frac{1}{2} \sum_i v_i^2$ is $(v_i - v_j) \sin(\theta_j - \theta_i)$; each update clips negative weights, hollows the diagonal and restores the total resource. This constrained update does not guarantee monotonic descent during joint noisy dynamics. Each seed runs four arms on a single phase-noise stream: the gradient step, a symmetric hollow random step rescaled to the Frobenius norm of that arm’s own gradient, that arm’s own gradient step composed with a random node relabelling drawn afresh at every update, and a frozen kernel. Each direction is computed from the arm’s own phases and couplings, so the controls follow the gradient arm’s recipe and no control receives its contemporaneous step; the step sizes the recipe produces are comparable rather than identical, since the arms’ kernels diverge. At zero learning rate all four arms reproduce one trajectory, which is the contract the shared stream is tested against.

The nonzero grand-mean frequency drives the collective coordinate, since symmetric interactions cancel in its mean drift. That cancellation fixes $\sum_i v_i$ across phase fields, so the objective cannot fall below $N\langle\omega\rangle^2/D = 990$ and the update can only equalize drift across nodes rather than remove it.

The objective is sampled at the update cadence, at the instant each update has just acted, so every objective mean in this subsection is a mean of that immediate post-update readout and not a time mean over the intervals between updates. On it the frozen arm sits at 1380.6063 to 1382.3448; the gradient arm’s second-half mean is 1110.5659 to 1116.6277, removing a fraction 0.6773 to 0.6914 of the headroom between the frozen arm and that floor. The random arm removes -0.0173 to -0.0103 of it, so the reduction is specific to the gradient direction and not to displacing a kernel of that step size at fixed resource. What that readout measures is the objective the update leaves behind it at the moment it

acts. Whether the objective stays below the frozen arm through the interval that follows is a separate measurement, reported in Section 9.3, and it is the one a claim of sustained minimization would need. The alignment statistics below are read from the kernel, which is constant between updates, so sampling them at the update cadence weights them exactly.

A step size set the same way is not a matched deformation. Fresh isotropic steps cancel across updates, so the random arm ends at a cumulative kernel norm growth of 1.2879 to 1.3056 against the gradient arm’s 12.4094 to 14.6958, and hardly enters the channel by which clipping and rescaling concentrate a fixed resource onto fewer edges. The permuted arm enters that channel instead. A node relabelling is an isometry of the matrix and a bijection of the edges, so it preserves the Frobenius norm of the arm’s own gradient and the whole multiset of its entries and changes only which edges receive them; the arm therefore takes a step of that gradient’s size and of its shape, uncorrelated with the frequency clusters by construction. It reaches a norm growth of 11.4185 to 16.3296, a range bracketing the gradient arm’s, and a second-half mean order of 0.9184 to 0.9189, which lies inside the gradient arm’s range. Its descent fraction is -0.3319 to -0.1607 : a step of a gradient’s own magnitude delivered to the wrong edges raises the post-update objective. What the two controls jointly support is therefore a comparison between arms whose step sizes are set by the same rule and whose observed deformation and coherence ranges overlap, not between arms equated by construction.

The learning rate is selected by a sweep from 0.001 to 0.4 on one seed, reported in the run’s `REPORT.md`: the post-update objective’s second-half mean falls from 1377.9075 at the smallest rate to a minimum at 0.2 and rises again to 1248.2272 at the largest. Rates past the minimum lose that reduction while still meeting the plateau criterion, so the criterion does not by itself select a rate and no convergence in the rate is claimed.

Second-half mean order ranges from 0.9167 to 0.9222, so coherence is maintained throughout. All seeds meet the declared plateau criterion: the last-quarter mean differs from the preceding quarter by at most 5%. The environmental template is the covariance of unit-variance latent signals shared within each frequency cluster, rescaled and hollowed to match the coupling resource; it is a diagnostic template, not an input to plasticity. Alignment is reported by two statistics unchanged under rescaling the kernel. The ratio of mean within-cluster to mean between-cluster coupling starts at 0.9824 to 0.9948 and ends at 0.3507 to 0.6135, against 0.9965 to 0.9997 for the random arm and 0.7290 to 1.2935 for the permuted one. The permuted range is wide and contains the no-preference line. Concentrating a fixed resource onto few edges makes a ratio of block means fluctuate, and with three clusters two thirds of the node pairs are between-cluster, so a dominant edge lands between more often than within and the typical value sits a little under one. The three permuted values nonetheless lie above the gradient arm’s three without overlap, at kernel deformations that overlap almost exactly. Against 2000 node relabellings drawn per run, the final kernel is further from the true template than from a fraction 0.9510 to 1.0000 of them. The frozen arm’s initial kernel already sits at 0.6435 to 0.9110, so the gradient arm is read against the frozen arm and not against the null median; the permuted arm, at the gradient arm’s deformation, reaches 0.2680 to 0.8410.

A second partition of the same nodes settles what the loss of within-cluster coupling is a loss against. The frequency partition is the structure the environment carries; an interleaved partition of the same three group sizes, built so that each group draws equally from all three clusters, carries no frequency information, and a kernel with only frequency structure is neutral under it up to a correction of order $1/N$. Neither partition enters the update. The gradient arm’s interleaved ratio runs from 0.8071 to 3.0223 across the three seeds, on both sides of the no-preference line, where its frequency ratio is 0.3507 to 0.6135 on all three. The interleaved partition is one member of a family, so the calibrated form of the same reading is to score the frequency partition against 2000 relabellings of itself, each keeping the group sizes and losing the correspondence with frequency: at most a fraction 0.0495 of them give a lower ratio than the frequency partition does, on every seed, against 0.0755 to 0.3480 for the frozen arm’s untouched kernel. Concentrating a fixed resource onto few edges makes any such ratio fluctuate widely, which is why the interleaved value alone is uninformative and the percentile against the family is not.

Two readouts are reported and not relied on. The initial-minus-final Frobenius distance to the template lies in $[-58.0962, -48.1760]$, but the kernel norm grows by a factor 12.4094 to 14.6958 over the same interval, and a fixed total resource constrains the sum of the entries rather than their norm; most of the distance growth is therefore concentration of mass onto fewer edges rather than loss of alignment, and the distance to the shuffled template grows with it. The off-diagonal correlation with the template ends at -0.0362 to -0.0208 , close to its starting value: at the sparsity the update produces, edge-level variance dominates a block-mean contrast that the within-over-between ratio resolves.

Minimizing squared drift under a conserved mean drift equalizes drift across nodes, which rewards

coupling between clusters of unlike frequency, whereas the template rewards coupling within clusters of like frequency. The disagreement is a property of this pairing of objective and template. The blind-partition reading is what separates that from the weaker statement it would otherwise be confused with: the update does not degrade whatever partition a reader scores it against, but locates the one partition the environment carries and moves against it. Which structure a different objective would move against, or whether any objective of this family aligns with an environment, is not settled by one pairing and is not claimed. The squared-drift objective includes the drive and fluctuating relative drift; it is not a thermodynamic entropy-production estimate, which would require probability-current information. The four panels of Figure 4 in the main article are these trajectories: the dissipation function, smoothed over 21 saved samples, against the $N\langle\omega\rangle^2/D$ floor; the order parameter against the $1/\sqrt{N}$ reference; the within-over-between coupling ratio against the no-preference line at unity; and the distances to the cluster template and to its node-label shuffle, which grow together as the kernel norm grows. Only the gradient arm lowers the post-update objective, and only it loses cluster alignment, the permuted arm holding the same deformation without either; these diagnostics provide no evidence of structural resonance or a thermodynamic entropy-production rate.

These finite trajectories establish neither stationarity of the adaptive process nor noiseless phase locking, and do not discharge the E45 modelling assumption or extend the fixed-phase descent theorem. What they exhibit is a reduction at the sampled instant together with a loss of alignment, and not a system that holds the objective below its frozen counterpart while it runs.

9.3 Sustained reduction over complete update intervals

A reduction at the sampled instant and a reduction maintained between updates are different quantities, and the second is the one a claim that a coupling update minimizes the objective needs. A bounded study measures it, on a design fixed before the runs were made and recorded in the repository at `tasks/f5_f6_design.md`, whose digest the saved summary carries.

The substrate is the one above: $N = 100$, $D = 0.1$, total coupling 400, physical duration 400, and the same initialization, frequencies, projection and objective. Only the learning rate and the physical interval between updates vary, over the product of rates $\{0.01, 0.05, 0.2\}$ and intervals $\{0.05, 0.5\}$ time units, each against a frozen arm on the same initial state and phase-noise stream. Objective and order are recorded at every integration step, on the state that drives that step, and averaged within complete update intervals; the terminal endpoint carries no duration and is excluded. The two assessment windows are the third and fourth quarters of the run.

The criteria are declared in advance and conjunctive: each window’s objective mean at least 5% below the paired frozen arm, each window’s mean order at least 0.8, a second-half within-over-between coupling ratio at most 0.8 and at least 0.1 below the frozen arm’s, and the final frequency partition at percentile at most 0.05 among 2000 size-preserving relabellings. Selection among the cases meeting the objective and coherence criteria is on the worst quarter’s objective ratio alone and is never shown an alignment value; a selected case would then have to meet every criterion on three held-out seeds at two integration steps, at fixed physical duration and cadence.

Table S2: Every case of the declared grid, each against its paired frozen arm. The objective columns are complete-interval means over the third and fourth quarters divided by the frozen arm’s; the order column is the smaller of the two windows’ mean order; the last two are the second-half within-over-between coupling ratio and the percentile of the frequency partition among 2000 size-preserving relabellings of the final kernel. The criteria are both ratios at most 0.95, order at least 0.8, and the alignment conditions of the text, all together.

Rate	Interval	Q3 ratio	Q4 ratio	Order	Within/between	Percentile
0.01	0.05	0.9974	0.9973	0.955	0.745	0.0090
0.01	0.5	0.9972	0.9969	0.951	0.770	0.0145
0.05	0.05	0.9878	0.9864	0.850	0.462	0.0060
0.05	0.5	0.9917	0.9934	0.902	0.548	0.0765
0.2	0.05	0.9924	1.0135	0.836	0.223	0.0560
0.2	0.5	0.9980	1.0032	0.923	0.582	0.2690

Of the 6 cases of Table S2, 0 meet the objective and coherence criteria together, so the design selects no candidate and the held-out confirmation is not run. The smallest worst-quarter objective ratio falls at rate 0.05 and interval 0.05, whose two windows sit 1.22% and 1.36% below the frozen arm against

the 5% the criterion asks for, at mean order 0.850 to 0.879, a within-over-between ratio of 0.462 and a blind-partition percentile of 0.006. Coherence and loss of alignment appear together across this grid; a sustained reduction of the declared size does not accompany them anywhere in it.

The grid contains the rate and cadence of the preceding subsection’s run, and in that case the two readouts separate: the post-update second-half mean is 1118.95 where the complete-interval mean is 1383.87 against the frozen arm’s 1383.06, a ratio of 0.998 in one window and 1.003 in the other. What the update takes off the objective at the instant it acts, the interval that follows returns.

The stop rule belongs to the design: the grid is not widened and no second candidate is tried once the outcomes are in. The study reports that this bounded family does not supply a sustained reduction, coherence and loss of alignment together. That is not a proof that no declared regime supplies them, and it is a statement about this squared-drift objective rather than about thermodynamic entropy production.

10 The cortical field identification

The framework identifies its posited continuum coupling kernel with the endogenous electromagnetic field of mammalian cortex. This is a claim about biology, not a consequence of any theorem above it.

10.1 Supporting evidence

Neurons do not only signal through synapses: transmembrane currents generate extracellular fields that shift membrane potentials of neighbouring cells, a mechanism known as ephaptic coupling. Endogenous fields of 1–5 mV/mm entrain spike timing Fröhlich & McCormick (2010); Anastassiou & Koch (2015), and stable electric fields carry information that survives representational drift in the underlying spiking population Pinotsis & Miller (2022, 2023a,b). The field is spatially continuous and superposes contributions from large populations — a medium in which spatially separated activity is combined without further mechanism. This is the integrative role the CEMI field theory assigns it McFadden (2002, 2020); Pockett (2012), physically supported by astrocytic syncytia joined by gap junctions Verkhratsky & Nedergaard (2018).

These observations support a modulatory interaction, but do not determine its integrated strength relative to phase diffusion. Continuity does not itself establish compatible local states or conscious unity. The model requires independent calibration of the coupling kernel, its spatial aggregation and its rate scale.

10.2 Why the rate calibration is not derivable

The conversion $K_{\text{eff}} = \gamma N s E f$ carries its whole physical content in γ , and no rearrangement of the remaining factors supplies it. The frustration control above records the dimensional half of this: s is a time shift per unit field, so sE is a time, and multiplying by an oscillation frequency leaves a pure number — the fractional phase advance a cell receives per cycle from the population field. Multiplying by a population size leaves it a pure number. A coupling in the sense of the main text’s stochastic model is an inverse time, and γ is the entire gap between the two.

The substantive half is that γ is a hypothesis about persistence, not a unit conversion to be fixed by convention. It states how a per-cycle phase advance accumulates into a drift rate, which depends on how long a field-driven timing shift survives. Three candidates are available and they are not interchangeable: $\gamma = f$ takes each cycle’s advance to be retained in full and independently; $\gamma = 1/\tau$ for a phase relaxation time takes it to decay as the medium relaxes; $\gamma = D$ takes it to accumulate at the rate phase itself diffuses. These need not agree, in cortex or anywhere else, and the field measurement chooses between them only if the persistence of induced timing shifts is measured alongside it.

The third candidate is a physical hypothesis, not an algebraic error. Under $\gamma = D$ the threshold test $K_{\text{eff}}/D > 2$ reduces to $N s E f > 2$; that cancellation is legitimate if an independent experiment establishes that persistence rate. It is unsafe to choose $\gamma = D$ merely because it cancels the unmeasured diffusion, or to recover any γ by requiring the model to cross its own threshold or fit the Bessel relation. Identification requires coupling, diffusion and the persistence rule to be estimated in one cortical regime by routes independent of the field arithmetic, which is item 1 of the main text’s protocol.

10.3 Finite-sheet spatial-decay control

A finite stochastic control asks whether fixing total coupling while shortening its spatial reach fragments the phase field. On a periodic 128×128 sheet spanning 2.0 mm, an exponential kernel was row-normalised to total coupling $K_0 = 8$ at every decay length; the dynamics used thermodynamic diffusion $0.5 \text{ rad}^2/\text{s}$, quenched Gaussian frequency spread 1.5 rad/s , and 20,000 Euler–Maruyama steps per length. With steady $r > 0.2$ as the declared operational coherence criterion, the smallest interpolated decay length above which every larger sampled length remained coherent was 0.0140 mm. On this grid that is 0.90 lattice spacings, and the shortest sampled length from which coherence is sustained is 0.94 spacings, so the sheet places the boundary below its own resolution.

Rerunning the transition band and the lengths above it on a 256×256 sheet of the same extent — half the spacing, so a given physical length spans twice as many cells — leaves the boundary at 0.01413 mm, 1.01 times its coarse value and 1.81 spacings on the finer grid. The operational crossing is thus nearly unchanged under this one refinement. Two threshold interpolants agreeing is not convergence of the dynamics they interpolate between: at 0.01467 mm, the shortest sampled length from which both sheets stay coherent, steady order is 0.6176 on the coarse sheet against 0.2687 on the finer one, and the refined crossing is still only 1.81 cells wide. The transition-local trajectories are metastable and non-monotone on both sheets, so that number is an operational boundary rather than a critical-point estimate with statistical precision.

At the empirical controls 0.1, 0.2 and 0.3 mm, steady r was respectively 0.9424, 0.9427 and 0.9427, and winding-defect density was at most 2.4×10^{-6} . From 0.1 mm out to the sweep’s longest range 8.0 mm, which is 4.0times the sheet’s extent, steady order spans only 0.00038, so the empirical band sits inside a saturated plateau rather than near a boundary. Thus this finite, single-seed parameter sweep does not show fragmentation in the empirical decay-length band. It neither measures cortex nor proves a transition, and it leaves seed dependence and the near-boundary metastability unresolved.

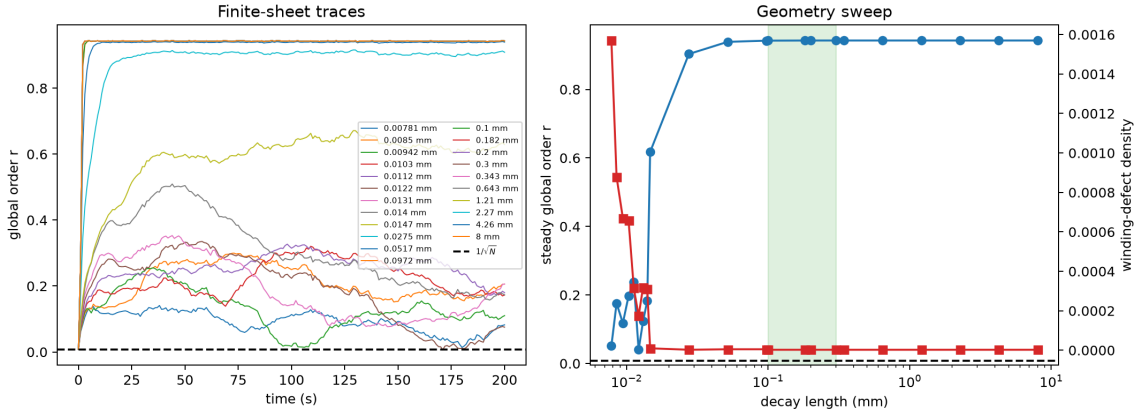


Figure S3: Finite-sheet heuristic evidence under a row-normalised spatial kernel. The left panel shows global order traces and the $1/\sqrt{N}$ finite-size floor; the right compares steady order and wrapped plaquette winding-defect density across physical decay lengths. The shaded band is 0.1–0.3 mm. This sampled simulation establishes no general transition or biological identification.

10.4 Critics

The longest-standing objection is that the fields are too weak to matter, pressed from within the field-theoretic camp by Pockett herself Pockett (2012, 2002). Vöröslakos and colleagues found that applied current is shunted through scalp, delivering fields at or below 1 V/m in cortex Vöröslakos et al. (2018). That is a bound on *applied* rather than endogenous fields, but it constrains how much causal work a field of that magnitude can do, and is the reason we do not claim the field drives spiking.

10.5 Falsification conditions

The physical identification would be challenged if independently calibrated field-mediated coupling were insufficient for the proposed regime, or if a selective perturbation of that interaction left the proposed

neural and behavioral endpoints unchanged. Such a perturbation requires controls for direct effects on transmembrane currents and other coupling pathways; selectivity is not assumed experimentally available. Recovery tests must distinguish the coupling-gated account from other slow mechanisms under the same observation model. Coherence alone is not equated with report or experience, so dissociations must be evaluated against independently specified endpoints and the additional compatibility and reflexivity hypotheses.

11 Relationship to other theories

Integrated information and global-workspace theories supply different descriptions of integration and access Tononi et al. (2016); Baars (1997); Dehaene et al. (2014). The present model specifies a candidate physical coupling and a consistency condition on local states. Gluing does not derive an exclusion principle, and continuity alone does not force compatible local descriptions. A digital architecture is not excluded by the sheaf construction. These mathematical differences do not settle which systems have conscious experience.

12 The sleep-inertia prediction: assumptions, alternatives, and falsifiability

12.1 Assumptions

The effective coupling must depend on extracellular geometry through a separately calibrated relation. The stationary ansatz assumes identical frequencies in a rotating frame and positive phase diffusion. The square-root temporal onset additionally requires a nonzero crossing speed and adiabatic branch tracking, which critical slowing down limits near threshold. The theorem does not prove dynamical selection. Finite-size fluctuations require an operational escape criterion above the fluctuation floor; neither an exactly flat observable foot nor an infinite initial slope is guaranteed.

12.2 Finite-ramp numerical check

A finite- N Euler–Maruyama ensemble quantifies the range over which the adiabatic reading is accurate. With identical frequencies, $D = 1$, $N = 2000$, 32 seeded replicas and $\dot{K} = v \in \{10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}\}$, escape is defined as the first of three consecutive sampled points at $r \geq 0.2$ after K_c . The fastest leg is right-censored: only 5/32 replicas escape before $K = 2.5$. All replicas escape at the three slower speeds, with mean delays $\Delta K = 0.1947, 0.0684$ and 0.0253 . Their log–log fit gives $\Delta K \propto v^{0.443}$, near the predicted exponent $1/2$ at this resolution. At fixed $v = 10^{-2}$ the delay is monotonic in N , increasing from 0.0619 through 0.1947 to 0.2341 as N increases from 500 through 2000 to 8000. The corresponding ratios $\Delta K/\sqrt{\ln N}$ are 0.0248, 0.0706 and 0.0781, which do not collapse to a size-independent constant. At $N = 500$, individual replicas reach $r = 0.436$ before K_c , while at $N = 8000$ they reach only 0.106; the absolute criterion $r \geq 0.2$ therefore fires almost immediately at the smallest size. The control demonstrates the protocol hazard warned of in the main text’s measurement protocol: a fixed absolute criterion meeting a size-dependent critical fluctuation floor.

The early-foot fit $r = A(K - K_0)^{\beta_{\text{eff}}}$ is declared over the order window $0.1 \leq r \leq 0.4$. Its exponents are read against the same estimator applied to the exact stationary branch on each leg’s own coupling grid, which returns $\beta_{\text{eff}} = 0.443, 0.443, 0.443, 0.448$ from fastest to slowest: the window reaches far enough past threshold that the estimator falls short of $1/2$ even on a branch whose exponent is exactly $1/2$, and that shortfall belongs to the window rather than to the drive. Against that reference the drive still dominates at $v = 10^{-2}$, where $\beta_{\text{eff}} = 0.898$ stands well above the adiabatic value; the two slower legs return 0.471 and 0.417, which fall on either side of their references rather than approaching them from one direction. The legs realise different windows and are comparable only so far as those agree: $K - K_c \in [0.04, 0.31]$, $[0.01, 0.19]$ and $[0.00, 0.20]$ over 28, 19 and 200 samples. At the fastest ramp $v = 10^{-1}$ the ensemble mean never reaches the declared window within the simulated interval: the fit truncates to $r \in [0.100, 0.151]$ at $K - K_c \in [0.32, 0.50]$ over 19 samples, so $\beta_{\text{eff}} = 0.964$ is the local log-slope of a strongly delayed trajectory rather than an onset exponent. The shift K_0 is returned at its lower bound on 4 of the 4 legs, so pinning describes the estimator on this family of traces and does not single out the fastest fit. A mapping from these dimensionless rates to human recovery times requires an independently calibrated phase diffusion and coupling trajectory. The independently estimated (a, r)

traces remain close to $I_1(a)/I_0(a)$ even where the trajectory lags the stationary branch. Post-critical RMS deviations are 0.00245, 0.00599, 0.00689, 0.00705; using the slowest run as the finite- N floor Dev_0 , none reaches $2\text{Dev}_0 = 0.01410$. The specified critical speed v^* is therefore not bracketed by this sweep. This numerical check exhibits finite sampled dynamics only: it proves neither stability nor dynamical selection and does not discharge the adiabatic assumption.

12.3 Alternative models and failure conditions

Exponential relaxation is one comparator, not the entire class of electrical models. Slow latent variables, observation filtering and finite-size effects can also generate delayed or sigmoidal traces. Comparisons should use common preprocessing and observation models, with onset criteria specified before inspecting outcomes. A linear apparent onset counts against the adiabatic branch interpretation only in a regime where its assumptions hold.

The concentration–coherence relation tests the stationary-density model. Its failure over an informative concentration range constrains that model; its success does not identify the coupling medium. The ratio a/r is a consequence of the same density ansatz and is unreliable near the fluctuation floor. A coupling–geometry association requires an independently validated geometry proxy and an independently defined recovery endpoint. Stationary absence of a fold does not rule out finite-rate hysteresis.

12.4 What successive stationary phase distributions cannot separate

The first item of the protocol asks for K and D in the same units because the stationary density does not carry them separately. In the identical-frequency rotating frame that density is von Mises with concentration $a = Kr/D$, so a window of equilibrated phases fixes the ratio K/D and nothing beyond it. Three recovery mechanisms therefore agree on every stationary observable, window by window. Write $s \in [0, 1]$ for a declared window coordinate, $q(s)$ for a schedule rising from 1.5 to an endpoint b , and $r(q)$ for the coherent branch of the stationary self-consistency relation, which is zero for $q \leq 2$. Increasing coupling is $K = q(s)$ at $D = 1$; decreasing diffusion is $K = b$ at $D = b/q(s)$; and a common drive is oscillators with no coupling to each other, each relaxing towards a shared phase ψ under a prescribed $-h(s) \sin(\theta - \psi)$ at $D = 1$ with $h(s) = q(s)r(q(s))$. The first two share $K/D = q(s)$; the third has stationary concentration h/D , which is that same number. Their stationary distributions coincide window for window, so no estimator reading only those distributions separates them. The third is one observationally matched drive and not a claim about every drive.

An ideal feasibility check confirms that the coincidence is exact rather than merely close. Nine equally spaced windows, exact densities on a periodic grid of 4096 points, endpoints $b \in \{2.6, 3.2, 4\}$ and orientations $\psi \in \{0, 0.7\}$ give 18 cases. Concentration is estimated by regression of log density on the cosine and sine of phase and coherence independently by circular integration; each family is then fitted for b over the same range against the same predicted concentration curve, with the generating family, K , D and true b withheld from the fit. The three families return the same b on every case, agreeing to within 8.9×10^{-16} at squared errors of at most 5.2×10^{-14} , and 0 of the 18 cases identify a unique generator. No family meets the declared gate of unique recovery on 80% of its cases, and the study stops there: observation noise, filtering and pooling applied to distributions that are already identical cannot separate them.

The observable the protocol is missing is the physical clock. Coupling and diffusion scaled together leave every stationary density untouched and change the rate at which phase moves, so resolved increments carry what equilibrated windows discard: along an ideal continuous path the quadratic variation accumulates at rate $2D$, and conditional increment means constrain the drift. Separating an endogenous coupling from a common drive is not settled by the clock either, since in the mean-field description the two impose the same drift on each oscillator; that separation needs a measured drive or a controlled perturbation. Neither measurement is validated by this pilot, and neither would identify the coupling as electromagnetic. The model comparison of item 6 is accordingly a comparison of time courses under a declared observation model, and identifies the mechanism behind a recovery only to the extent that the observations reach past the stationary density.

13 Interpretive scope

The formal endpoint is the section glued on a specified compatible cover, fixed by a specified reflexive map. The interpretation developed in the main article identifies the section with conscious unity and its

accurate self-representation with a minimal self. Uniqueness is relative to the cover and map; identifying these with meaningful neural descriptions and a neural self-encoding requires independent evidence. Restriction-resonance supplies the relation between the encoding and the state it represents.

The full commitment proposes that unity and reflexivity occur together in conscious episodes. Gluing by itself does not require reflexivity, so a dissociation could challenge that requirement while leaving the unity clause assessable. The physical recovery tests concern coupling and phase dynamics. Testing the interpretive clauses also requires validated measures of compatibility and self-encoding, together with an independent experience endpoint.

14 What a measurement of overlap compatibility would require

The main article’s compatibility requirement needs a neural observable distinct from phase order. This section specifies that measurement: decoded local descriptions, restrictions to shared domains, uncertainty under decoding noise, and validation on examples with known compatibility. The specification is not implemented here.

14.1 Three components

A restriction map. Two regions with overlapping receptive territories may each support a decoder for content on the shared part. Compatibility is a relation between the resulting objects, so the decoder and its restriction map must be specified before any statistic is computed. The formalization uses finite spatial measures: on its discrete substrate two patch sections have the same restriction exactly when they assign the same mass to every shared site (`restrict_eq_iff_density0n_eq0n`). A content decoder using probability laws needs additional structure: equality of overlap marginals is necessary, but global extension and uniqueness must be tested or assumed separately rather than imported from the spatial-measure theorem.

A statistic with a null. A distance between the two restricted distributions is not yet a measurement, because decoding error alone makes that distance positive. The statistic needs the distribution of the distance under compatibility — what two decoders of the same underlying content produce given their own noise — so that incompatibility is a departure from a stated null and not a nonzero number. Overlapping territories supply dependent samples, and the uncertainty has to be computed accordingly.

Validation where the answer is built rather than inferred. The estimator must be exercised on data in which compatible and incompatible overlaps are constructed by design, so that its sensitivity is known before it is applied where the answer is not. The formalization supplies the configuration to target: local states at a single common phase whose densities disagree where the patches meet, which satisfies every other requirement of the gluing theorem and fails compatibility alone (`overlap_agreement_fails`). An estimator that cannot separate that configuration from its compatible counterpart at realistic decoding quality is not yet a measurement of the property the framework needs.

14.2 Three ways such a measurement fails silently

A decoder that shrinks both regional posteriors toward a shared prior reports compatibility whatever the regions encode, having replaced two local descriptions with one global one. A decoder carrying independent per-region bias reports incompatibility whatever the regions encode. These are opposite failures of the same component, neither is visible in the compatibility statistic itself, and both are why the constructed-answer validation above is a requirement rather than a refinement.

The third is circularity. A compatibility statistic computed from phases measures phase agreement, which the counterexample shows can coexist with incompatible local descriptions. E78 connects coherence to a supplied cover’s relaxation; the cover’s own compatibility field supplies overlap agreement. A statistic for that agreement must use the local descriptions and their restrictions.

14.3 Scope

A validated observable would make agreement between the chosen local descriptions measurable. Establishing that coherence maintains it would require a dynamical test in addition, and selecting the content model and cover would remain an independent task. A full test of the consciousness identification would also measure the proposed self-encoding and an experience endpoint: compatibility and supercritical coupling alone leave the reflexive clause unevaluated. Absence of report must be distinguished from

evidence against an episode. If compatibility cannot be distinguished from incompatibility at achievable decoding quality, the identification is outside the reach of that measurement.

15 Exploratory EEG exercise of the concentration–coherence estimators

This section reports an exploratory check of the parameter-free relation $r = I_1(a)/I_0(a)$ on public propofol-sedation EEG (OpenNeuro ds005620 Bajwa et al. (2025)). The saved summary and the generated macros record every numerical result below. This is an estimator exercise, not a test of awakening or of the electromagnetic-field commitment.

15.1 Preprocessing

For each recording block, 20 seconds of data and 3 seconds of padding on each side were downloaded from the public S3 bucket. Each channel was detrended, bandpass filtered with a fourth-order Butterworth filter using second-order sections (`sosfiltfilt`), and converted to instantaneous phase with the Hilbert transform. The padding is discarded after filtering in every montage. The primary analysis uses 61 adjacent bipolar pairs formed from the 62 scalp EEG channels; EOG and EMG channels are excluded.

15.2 (a, r) estimation

Each recording was divided into non-overlapping 100 ms bins (500 samples at 5 kHz). The estimand is the pooled spatiotemporal phase distribution over 62 sites and 100 ms, not an instantaneous 62-site spatial distribution. Pooling is required because the independent log-density estimator requires at least 100 phase samples. For each bin t :

- **Order parameter r :** $r_b = \left| \frac{1}{NM} \sum_{j=1}^N \sum_{m=1}^M e^{i\theta_j(t_{bm})} \right|$, where $N = 62$ channels and $M = 500$ samples per bin. This is the resultant length of the pooled sample, not an average of instantaneous spatial resultant lengths.
- **Concentration a :** The slope of the log-density of the same phase distribution regressed on $\cos \theta$, as specified in the main text. Phases are centred on their circular mean, binned into 40 bins on $[-\pi, \pi]$, and log of the binned density (with a 0.5 pseudocount) is regressed on $\cos \theta$ by *unweighted* ordinary least squares. Under the von Mises model $\log \rho(\theta) = a \cos \theta - \log(2\pi I_0(a))$, so the slope is a .

The estimator must not be a maximum-likelihood one. The von Mises family is a one-parameter exponential family whose sufficient statistic is $\sum_j \cos \theta_j$ — that is, r itself — so any ML estimate of the concentration solves its score equation $I_1(\hat{a})/I_0(\hat{a}) = r$ and returns $\hat{a} = A^{-1}(r)$ identically. The pair $(\hat{a}, r)$ would then lie on $r = I_1(a)/I_0(a)$ for any input whatsoever, including phases drawn from a distribution that is not von Mises, and the collapse would be an algebraic identity rather than a test. The implemented regression is unweighted, so it uses the log-density shape rather than defining concentration by inversion of the first trigonometric moment. Other weighting rules are different estimators and are not assumed equivalent to maximum likelihood.

The pseudocount biases the estimator, so its null residual must be calibrated at the sample size and dependence structure of the intended measurement. The exploratory bins contain 30500 phase values, but the proposed spatial protocol permits only 100 sites. At 100 independent samples, 500 von Mises draws with true $a = 2.0$ return mean $\hat{a} = 1.346$ and mean residual $+0.146$ with standard deviation 0.028. A large residual at that count can therefore be estimator bias rather than evidence against the density relation. Independence in this calibration is itself optimistic for EEG; a spatial experiment must use a dependence-aware null.

The estimator also separates synthetic non-von-Mises alternatives from correctly sampled high-count von Mises data. This confirms that the choice of unweighted log-density regression avoids the maximum-likelihood tautology; it does not calibrate a low-count or dependent neural measurement.

15.3 Bootstrap confidence intervals

95% confidence intervals on the mean (a , r) per block were obtained by bootstrap resampling of the time bins (2000 resamples, percentile method). This procedure does not preserve dependence between neighboring bins; its intervals are exploratory rather than a calibrated time-series uncertainty estimate.

15.4 Narrowband analysis

The pipeline was run separately on theta (4–8 Hz), alpha (8–12 Hz), beta (13–30 Hz), gamma (30–40 Hz), and broad (4–40 Hz) bands.

15.5 Results

Residuals are evaluated per 100-ms bin and then averaged, rather than computed from block-mean a and block-mean r ; because I_1/I_0 is nonlinear, the latter is not the mean residual. Figure S4 shows that this particular pooled estimator is sensitive to very short windows. It does not validate an instantaneous spatial estimator.

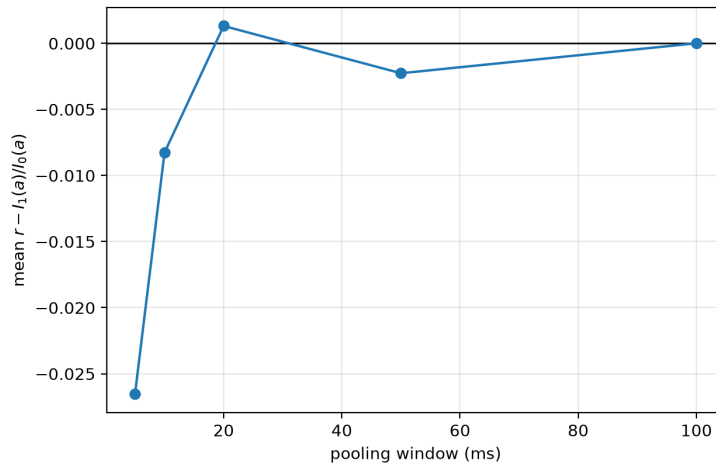


Figure S4: Sensitivity of the mean collapse residual to the spatiotemporal pooling window. This assesses the pooled estimator and does not turn 62-channel EEG into an instantaneous spatial sample.

- **Cross-subject bipolar summary.** Across 8 sedated subjects, the mean concentration is 0.104 (range 0.087–0.142) and mean order is 0.052. The mean residual is -0.0002, with RMSE 0.0027 and maximum absolute residual 0.0364.
- **Narrowband.** The theta, alpha, beta, gamma and broad-band mean residuals are -0.0018, +0.0001, +0.0001, +0.0000 and +0.0000, respectively, for sub-1016, sed run-1, bipolar.

Reference choice remains a material analysis decision. The corrected summary fixes the primary montage and padding rule, but does not establish a mechanism for differences between montages.

15.6 Interpretation

The exploratory bipolar data are compatible with the relation at their observed range and pooling choice. This is a necessary condition for the stationary-density model, not evidence for an electromagnetic coupling medium or a test of the awakening prediction.

The observed range is still too narrow to discriminate the Bessel ratio from its linear approximation at the trace's scatter. The curve departs from that approximation by at most 0.0095 over the observed concentration range; a discriminating awakening experiment must span concentrations around 1.5, where the separation is 0.154, and must calibrate its estimator at the actual spatial sample count.

The result does not test the empirical commitment that the endogenous electromagnetic field is the coupling medium, nor the stipulation that the global section is experience. The coupling-gated awakening prediction remains untested here; these recordings concern propofol sedation rather than recovery from sleep.

References

- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable electric fields emerge from and allow representational drift. *NeuroImage*, 253, 119058.
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.
- Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the Mind. *Oxford University Press*.
- Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25, 76-84.
- Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature Reviews Neuroscience*, 11(2), 127-138.
- Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How particular is the physics of the free energy principle?. *Physics of Life Reviews*, 40, 24-50.
- Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some parts of the free energy principle. *Entropy*, 23(3), 293.
- Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Emperor’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.
- Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows memory network formation. *Cerebral Cortex*, 33(17), 9877-9890.
- Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling: Electric fields sculpt neural activity and "tune" the brain’s infrastructure. *Progress in Neurobiology*, 226, 102465.
- McFadden, J. (2002). The conscious electromagnetic information (cemi) field theory: the hard problem made easy?. *Journal of Consciousness Studies*, 9(8), 45-60.
- Pockett, S. (2012). The electromagnetic field theory of consciousness: a testable hypothesis about the characteristics of conscious as opposed to non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.
- McFadden, J. (2020). Integrating information in the brain’s EM field: the cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1), niaa016.
- Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may guide neocortical network activity. *Neuron*, 67(1), 129-143.
- Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous electric field activity: why bother?. *Current Opinion in Neurobiology*, 31, 95-103.
- Pockett, S. (2002). Difficulties with the electromagnetic field theory of consciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- Verkhatsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physiological Reviews*, 98(1), 239-389.
- Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Correa, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced unconsciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J. F., & Juel, B. E. (2025). A repeated awakening study exploring the capacity of complexity measures to capture dreaming during propofol sedation. *Scientific Reports*, 15, 32012.
- Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... & Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult brain. *Science*, 342(6156), 373-377.

- Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space. *Physiological Reviews*, 88(4), 1277-1340.
- Barbour, B. (2017). Analysis of claims that the brain extracellular impedance is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.
- Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return of Maxwell's demon. *Studies in History and Philosophy of Science Part B: Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.
- Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3), 183-191.
- Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal of Physics A: Mathematical and General*, 9(8), 1387-1398.
- Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.
- Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics of prediction. *Physical Review Letters*, 109(12), 120604.
- Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation: The phase-space perspective. *Physical Review Letters*, 98(8), 080602.
- Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.031015.
- Ito, S., & Sagawa, T. (2013). Information thermodynamics on causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/PhysRevLett.111.180603.
- Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L. (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-181.
- Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-archive.pitt.edu/115.
- Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen, H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics, vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork bifurcations with additive noise. *Probability Theory and Related Fields*, 122, 341–388. doi:10.1007/s004400100174.
- Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13, 113036. doi:10.1088/1367-2630/13/11/113036.
- Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin, P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11), 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.
- Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave patterns underlie complex brain dynamics and are related to cognitive processing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-023-01626-5.